

Cognitive computing technological trends and future research directions in healthcare — A systematic literature review

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ABSTRACT

Background and aim: Cognitive Computing systems are the intelligent systems that thinks, understands and augments the capabilities of human brain by blending the technologies of Artificial Intelligence, Machine Learning and Natural Language Processing. In recent days, maintenance or enhancement of health by preclusion, prognosis, and analysis of diseases has become a challenging task. The increasing diseases and its causes becomes a big question before humanity. Limited risk analysis, meticulous training process, and automated critical decision-making are some of the issues of cognitive computing. To overcome this issue, cognitive computing in healthcare works like a medical prodigy which anticipates the disease or illness of the human being and helps the doctors with technological facts to take the timely action. The main aim of this survey article is to explore the present and futuristic technological trends of cognitive computing in healthcare. In this work, different cognitive computing applications are reviewed, and the best application is recommended to the clinicians. Based on this recommendation, the clinicians are able to monitor and analyze the physical health of patients.

Methods: This article presents the systematic literature on the different aspects of cognitive computing in healthcare. Nearly seven online databases such as SCOPUS, IEEE Xplore, Google Scholar, DBLP, Web of Science, Springer and PubMed were screened and the published articles related to cognitive computing in healthcare is collected from 2014 to 2021. In total, 75 articles were selected, examined and their pros and cons are analyzed. The analysis is done with respect to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.

Results: The basic findings of this review article and their significance for theory and practice are mindmaps portraying the cognitive computing platforms, cognitive applications in healthcare, and use cases of cognitive computing in healthcare. A detailed discussion section highlighting the present issues, future research directions and recent applications of cognitive computing in healthcare. Accuracy analysis of different cognitive systems conclude that the Medical Sieve achieves 0.95 and Watson For Oncology (WFO) achieves 0.93 and hence proves to be the prominent computing systems for healthcare.

Conclusions: Cognitive computing, an evolving technology in healthcare augments the clinical thought process and enable the doctors to make the right diagnosis and preserve the patient's health in good condition. These systems provides timely care, optimal and cost-effective treatment. This article provides an extensive survey of the importance of cognitive computing in the health sector by highlighting the platforms, techniques, tools, algorithms, applications, and use cases. This survey also explores about the works in the literature on present issues and proposes the future research directions of applying cognitive systems in healthcare.

1. Introduction

Cognitive computing simulates human thoughts, imitates human intelligence, [1] and builds up an automated computerized human model. Cognitive computing ingests reasoning abilities into machines to act similar to humans. Cognitive science is a multidisciplinary field which explores the mental ability of human beings. The cognitive

behavior [2] of human beings is a two-staged closed-loop technique. In the initial stage, humans become aware of the physical environment through their sense organs and make a decision.

In the next stage, this decision serves as the input, and the brain processes and performs the actions. Thus, cognitive computing involves streaming of structured and unstructured big text data or smart text

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data by applying Natural Language Processing (NLP), Machine Learning (ML), and Artificial Intelligence (AI) algorithms for reasoning, learning, predicting, and intelligent decision making. These cognitive computing systems implements model-based reasoning capabilities to perform understanding, reasoning, and thinking. The cognitive computing technologies goes beyond next level of computing [3] where different rules or constraints are framed using knowledge engineering methodologies for analyzing the critical information and taking optimal decisions. The advanced applications of cognitive computing are highlighted. The features of cognitive systems¹ are (i) Adaptive(dynamic decision making), (ii) High interaction (good interactive interface and NLP analysis), (iii) Repetitious and stateful (gathers relevant inputs and saves state), (iv) Contextual (current context analysis)

Cognitive computing systems outlooks the human mind as an information processing system [4] which uses various algorithms for knowledge representation and reasoning. These systems works by drawing inferences and conclusions using different algorithms. Cognitive systems are agent based systems which are adaptive that can model the cognitive tasks. The behavior that defines cognitive thought processes is known as cognitive task [5]. Assessments of cognitive abilities and brain activity are conducted using specific cognitive tasks. Memory, language, social, decision-making, and sensorimotor skills are frequently used in cognitive tasks. The cognitive task is differed from traditional task in which traditional task concentrates on explicit observable behaviors that support observable task performance, whereas cognitive task is focused on the psychological processes that encompass the behavior.

Cognitive computing systems [6] helps in taking proactive decisions by assessing the inadequate amount of data, discovering evidence-based hypothesis and scoring it. These systems replicates the human reasoning capabilities [7] and with massive amounts of data, the system learns more and becomes more precise and efficient. Cognitive systems are not involved in programming but these are dynamic and self learning which takes in large amounts of past experience and renders successful actions and decisions. Cognitive computing systems are self learning systems because data mining techniques, pattern recognition, natural language, and human senses are incorporated as well as the refinement of systems based on real-time data acquisition from patients and other sources. Essentially, these systems work in a similar way to the human brain and learn on their own.

Cognitive computing uses automated technologies [8] for acquiring relevant relationships from the data, understands and interprets the data, extracts useful patterns from the data, generates inferences and draws conclusions. Cognitive computing technology fuses many different technologies such as AI, ML, NLP to recognize human cognitive capabilities and increase the efficiency and productivity of systems. Main characteristic features of cognitive systems include memorization, continuous and experience learning, recall, complex reasoning, strong comprehension, logical thinking, deep insight extraction, expert training, model optimization, personalized analysis, self adaptation, dynamic systems, contextualize knowledge and so on.

The ultimate goal of cognitive computing technology [9] deals with the development of optimal automated systems that not only solves problems but also suggests many alternatives without human intervention. Cognitive computing systems handles large collection of existing historical data for analysis and applies intelligent algorithms to determine the relations from the data and generates proactive, intuitive and decisive results. Cognitive systems thinks, understands the data [10] by naturally interacting with humans and applies intelligence. Some features of cognitive computing systems includes sensing, interacting, thinking, understanding, dynamic, context aware, reasoning, interpreting, action taking, evidence based decision making, deeper level comprehension, generating insights and so on.

Cognitive systems provides the pathway for the machines to fuse reasoning capabilities [11] similar to humans. Cognitive computing is an interdisciplinary field of research with various ML-based intelligent services which riggings computational intelligence in various real-world applications, such as healthcare, cognitive-based learning systems, cognitive robotics, retail, security, smart cities, finance, cognitive search engines, intelligent transportation systems, cognitive gaming, telecommunications, agriculture, real estate, public sector, media and so on. Healthcare, which is one of India's largest sectors [12], mainly deals with the preservation or enhancement of health through proper prognosis and treatment. The increasing cost for the treatment of diseases leads to the threat in doctor to patient relationships. Cognitive computing technologies emerged in healthcare to help medical professionals for the early assessment and prevention of the diseases. In the healthcare field, cognitive computing technology has a high impact. This paper surveys recent developments in healthcare using cognitive computing.

Cognitive systems provide specific details about the patient's health to the doctors for easy diagnosis and right treatment. HealthIT Analytics, an online resource portrays that Google applies deep learning skills and Fast Healthcare Interoperability Resources (FHIR) standards to solve the problems of predictive analytics. Since healthcare data is multitudinous and complex, cognitive computing has an incredible impact on the health industry. In a lifetime, an average human generates nearly 1100 Terabytes (TB) of extrinsic data, six TB of genomic data, and 0.4 TB of clinical data. So, only 20 percent of the data are evidence-based, and the remaining 80 percent of the healthcare data generated by humans are unstructured. It has been estimated that by 2020, the amount of medical data [13] doubles every 73 days, which results in large volumes of unstructured data. Nearly 700,000 new articles are published by the medical journals every year. Improved patient health at lower cost is a benefit of cognitive healthcare decisions. The growth of data has been increased to a larger extent but extracting meaningful information from the large amounts of data has proven to be a challenging task [14]. So, the cognitive computing technology plays an important role in processing the data.

Cognitive computing processes large volumes of unstructured data [15] like clinical notes, journal articles, medical reports, operation notes, discharge summaries, and medical images to extract knowledgeable insights. According to the study in 2016 given by Grand View Research, Inc., the cognitive computing market in the fields of healthcare will be higher than five billion dollars by 2022. Thus, cognitive computing evolves in the field of medicine and healthcare. Generally, the cognitive systems involves the three fundamental processes [16] such as training, autonomous and continuous learning, and hypothesis generation. These systems analyzes healthcare data, derives the correlations, identifies hypothesis for the relations and generates the evidence based treatment options.

The cognitive computing systems [17] helps the medical scientists with reliable and meaningful information derived from the patient's medical records. Cognitive computing technology paradigm processes a large collection of data which can help healthcare professionals. This technology paradigm discovers different correlations by analyzing huge amounts of healthcare data, renders decisions by creating and scoring evidence based hypothesis. The first process involved in healthcare related cognitive systems are medical data collection and automated ingestion of all kinds of data. The second process is that whenever a user raises a question for example "What are the symptoms of a particular disease?". The third process deals with generating hypothesis wherein the system analyses the data and generates different symptoms as different hypothesis. The fourth process involves scoring the hypothesis by gathering the supporting evidences from the knowledge base. Finally, the optimal and best scored hypothesis is presented to the user.

To summarize, cognitive systems act as a decision support system to render reasonable insights from the large amounts of data. In short,

¹ <https://marutitech.com/cognitive-computing-features-scope-limitations/>

the cognitive systems renders best and optimal actions from the alternative list of actions. These systems replicates human brain capabilities by taking large amounts of data, performs correlation analysis, tests the hypothesis, makes inferences and derives conclusions. The fundamental steps include training, continuous and autonomous learning, dynamic knowledge base creation, hypothesis identification, evidence extraction, and confidence weighted score generation.

The common challenges faced by Researcher on handling Cognitive Computing systems are:

A cognitive computing system developer must collect enough relevant knowledge and represent it in a way that adds value to the system's knowledge pool. For capturing and representing healthcare knowledge, there must be experts who know the vocabularies and rules of healthcare [16]. Moreover, they need to explain how deep learning and machine learning are codified. Chemical symbols and medical images can be used in healthcare as a special language [18]. It is challenging to teach business-specific words and concepts, as well as systems structure and vocabulary.

The major contributions of this survey article are:

- (i) Depiction of cognitive computing process.
- (ii) Rendering schematic mindmaps to depict the cognitive computing platforms, cognitive healthcare applications, present issues and future enhancements of cognitive computing in healthcare, use cases of cognitive computing.
- (iii) Analysis of present challenges for cognitive computing in healthcare.
- (iv) Future enhancements for each challenge is proposed.

The rest of the paper is structured as follows Cognitive Computing Process section elucidates the process of cognitive computing, background section elucidates the background of cognitive computing technology. Related work section deals with the works related to cognitive computing in healthcare, and cognitive computing platforms section describe the various platforms. Cognitive tools and frameworks section explains the various tools and frameworks, and cognitive applications in healthcare section narrates the different applications, challenges and future directions. The next section outlines the use cases of cognitive computing in healthcare, discussion section elucidates the highlighted features of eminent cognitive systems, and the final section concludes the survey.

2. Cognitive computing process

The process of cognitive computing runs in sequential fashion and consist of seven major steps as shown in Fig. 1. The data collection step deals with the collection of large volumes of data in both structured and unstructured formats. The collected data is automatically ingested into the cognitive system. In the next step, a question is given as input and the system analyses the question and identifies the problem. The relation extraction step extracts the relations necessary for the problem from the cognitive knowledge base constructed by the system. The hypothesis generation step generates the possible answers for the questions based on the relations. The evidence extraction step extracts the evidence for each hypothesis. In the hypothesis scoring step, the hypothesis along with the evidence are scored and the evidence based hypothesis with high score are accepted and the evidence less hypothesis are refuted. Finally, the confidence score threshold is calculated and the best or optimal hypothesis is selected as the answer.

2.1. SLR map

The systematic literature review is divided into two phases wherein phase 1 explains in detail about the research domain cognitive computing and phase 2 describes about the research domain application i.e., cognitive computing in healthcare as shown in Fig. 2.

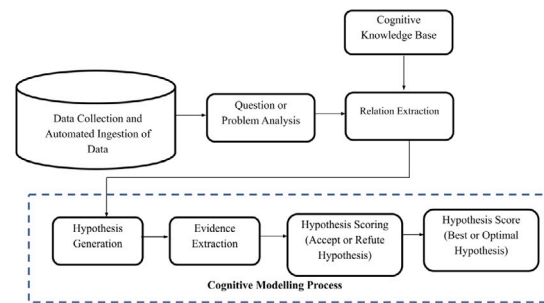


Fig. 1. Cognitive computing process.

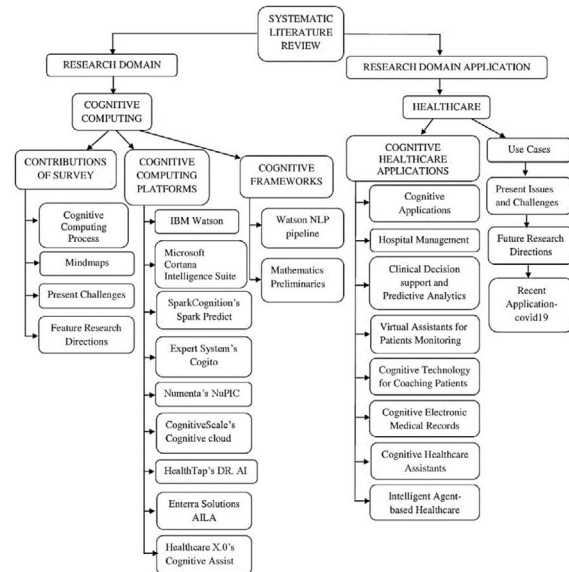


Fig. 2. SLR map.

3. Background

The word cognitive connects thinking and conscious thought process. In general, learning is the process of procuring new knowledge, etiquette, conscience, and so on. Cognitive learning mainly deals with acquiring the ability to solve problems with intelligence, thinking, and conscious thought. Cognitive assistant is an intelligent software agent which augments (making it full or adding or expanding to the fullest) human intelligence. Thus, cognitive computing is an intellectual discipline for simulating human thoughts and generating a computerized human model [19]. This model has NLP capability, experiential learning capability, natural human interaction, and critical decision making.

The cognitive computing technology reproduces human information processing capabilities. A human focused cognitive computing [20] has been designed in combination with the cloud computing technology. The proposed cognitive computing architecture incorporates IoT, big data analytics, and cloud computing. Some prominent cognitive technologies are reinforcement learning (reasoning out positive or negative rewards), deep learning (AI methods to mimic human brain). Many applications are analyzed such as creation of chatbots, opinion mining, face or emotion recognition and analysis, cognitive medical approaches and assessment of risk. The cognitive chatbots advancements and calculations is suggested as a future work. In perspective of sentiment analysis, cognitive AI or intellectual AI is used to draw conclusions and inferences.

Cognitive computing aims to replicate human thought processes in a virtual representation. The computer can imitate how the human brain

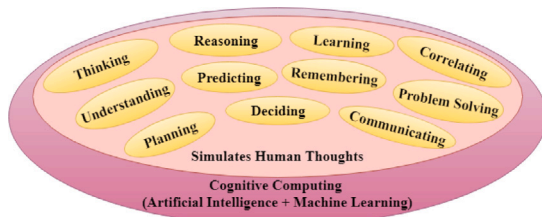


Fig. 3. Cognitive computing in relation to human thought process.

functions by using self-learning algorithms that use data mining, pattern recognition, and natural language processing. Cognitive computing blends the techniques of AI and ML to think analyze and understand the human capabilities. Cognitive computing resembles central cognitive activities such as thinking, reasoning, learning, correlating, understanding, predicting, remembering, problem-solving, planning, deciding, communicating, and other mental phenomena, as shown in Fig. 3. The cognitive machines combines the capabilities of knowing, reasoning, acting, perceiving, and autonomously reacting [21].

Cognitive Computing Consortium (CCC) defines that the cognitive machines are adaptable, interactive, iterative, stateful, and contextual. These cognitive machines gradually progresses to reason the human's central cognitive capacities such as thinking, reasoning, predicting, learning, remembering, understanding, correlating, problem solving, communicating, deciding, planning and so on. So, these machines are fused with these capacities to resemble the intelligence of human mind.

The creation of actionable knowledge from data is the goal of cognitive systems. Due to the availability of a lot of relevant data, a lot of computing power, and the high value of the usable knowledge for many big businesses, cognitive computing has recently grown significantly. A cognitive system creates actionable knowledge through four stages of processing.

Cognitive computing technology uses four phases [22] such as internal and external source analysis, data linking, knowledge or intelligence extraction and model deployment. Phase 1 — Internal and external source analysis takes the structured (operational) and unstructured data (interactions, events) as input and tags the data as output. Phase 2 — data linking involves data ingestion, selection, arrangement, sampling and aggregation where the tagged data are fused into large graphs. Phase 3 — model building consists of two techniques such as Bayesian statistical modeling and Deep Neural Network (DNN) technique to extract intelligence. Phase 4 — model deployment deals with generating decisions, insights, reports and visualizations. The inference from the DNN is given as shown in Eq. (1)

$$y_{k+1} = h(W_k \times y_k + b_k) \quad (1)$$

where W_k indicates the weight matrices, b_k describes the bias vectors, $h(x)$ indicates the non-linear function to compute y_k vectors.

Cognitive computing framework interconnects big data analytics, deep learning, ML [23] and it requires massive amounts of data to replicate human brain capabilities. Reinforcement learning algorithm, deep learning algorithms, NLP, decision automation and reasoning, emotional intelligence are blended into cognitive computing architecture. Tensorflow, Theano, Torch, and Caffe [24] are the frequently used cognitive computing platforms. The main advantages of cognitive computing [25] involves customer engagement, security, productivity, reduced risks, and reduced cost consumption.

The cognitive computing technology observes, interprets, evaluates and takes decisions by mapping to the features of big data [26] namely volume, veracity, velocity, value and variety. The survey portrays the progression of data to information, information to knowledge and knowledge to wisdom. The cognitive computing architecture based on big data has been presented which illustrates the applications,

libraries, databases, tools, frameworks and infrastructure. A conceptual model linking big data and analytics and cognitive computing has been proposed. The conceptual model converts the world to the smarter world. The data is created from the world and big data technology is applied to generate the existing knowledge. After deriving knowledge data analytics, cognitive computing and ML technology is applied to generate new insights and knowledge. In future, cloud computing can be integrated with cognitive computing and big data for generating more useful insights.

Cognitive computing technology is a predictive (foresight), prescriptive (insight) analytics and makes use of AI which simulates human intelligence by machines. In general, predictive analytics analyzes the current and historical data and makes predictions about future events, whereas prescriptive analytics deals with intelligence and modeling to recommend the next course of action. In healthcare, predictive analytics deals with predicting and diagnosing the disease of the patient in advance. After the diagnosis of disease, prescriptive analytics recommends the informed decision to be taken about the patient's treatment to the doctors. AI is a technology that uses ML, NLP, reasoning, and human interactions through speech and vision. The idea of Artificial Intelligent agent is that machines can make better decisions on behalf of humans. The intelligent cognitive agent helps humans to make smarter decisions leveraging the machines. A cognitive analytics system analyses large data sets and gives structure to unstructured data using multiple analytical techniques. The medical sector is now beginning to employ cognitive analytics to match patients with the most effective therapies. Today's examples of cognitive analytics include IBM's Watson, Apple's Siri, and Microsoft's Cortana.

Cognitive assistants [27] explore how technology helps older people affected by physical or mental disabilities and chronic diseases. Cognitive computing makes use of biologically inspired neural networks in order to simulate human cognitive process [28] and functions. Cognitive systems perform specialized processing and large scale analysis of data. Cognitive computing helps in preclusion of diseases, personalized medical treatment for patients, disease status identification, disease progression modeling, dynamic treatment regime, and so on. For example, the cognitive computing based Clinical Trial Matching (CTM) system [29,30] makes use of NLP techniques to identify the cancer attributes from the patient's medical records. The CTM system acts as an assistant to the physicians for efficient and accurate trial matching. This CTM system also scans the patients in less time when compared to doctor scanning. Another application for CTM system [31] involves identifying tumor attributes from the health records of breast cancer patients. The CTM system rendered faster results to the oncologists by providing matched trials prior on the day before their visit for actionable decision making. Moreover, cognitive systems are cost-effective systems that help healthcare professionals in making improved decisions about care and treatment. Cognitive technologies analyze Electronic Health Records (EHR), Electronic Medical Records (EMR), and Personal Health Records (PHR) to help doctors with informed decisions.

The ultimate goal of this survey is to portray the importance of cognitive computing in the field of healthcare by performing an analysis of many different companies' cognitive products, tools, frameworks, applications, and use cases.

4. Collection of published research papers

4.1. Search strategy analysis

The titles and abstracts of the research papers published in the four online databases such as SCOPUS, IEEE Xplore, Google Scholar, DBLP were screened. The research papers published during the period 2014 to 2021 were collected and analyzed. Table 1 shows the different search keywords used in different databases.

Table 1
Search keywords for each database.

Database	Keywords
SCOPUS	“cognitive computing in healthcare” OR “cognitive healthcare applications”
IEEE Xplore	“cognitive computing in healthcare” OR “IBM Watson” OR “cognitive healthcare applications” OR “cognitive computing”
Google Scholar	“cognitive computing in healthcare” OR “IBM Watson” OR “cognitive healthcare applications” OR “cognitive computing”
DBLP	“cognitive computing in healthcare” OR “cognitive computing”
PubMed	“cognitive computing in healthcare” OR “cognitive computing”
Springer	“cognitive computing in healthcare” OR “cognitive computing”
ScienceDirect	“cognitive computing in healthcare” OR “cognitive healthcare applications”

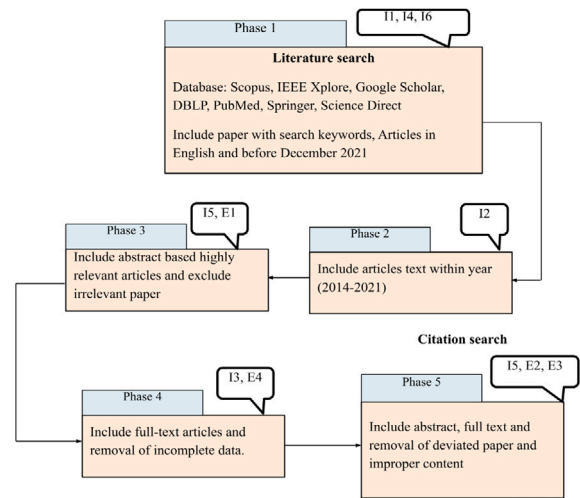


Fig. 4. Selection of research data.

Table 2
Inclusion criteria for the selection of research papers.

Inclusion criteria	Description
I1	Used keywords are cognitive computing in healthcare” OR “cognitive healthcare applications”. Operators of search syntax - OR. OR - at least one keyword must appear in the search query.
I2	Include papers published within 2014–2021
I3	Include full-text articles
I4	Include articles published in English
I5	Include abstract based highly relevant articles.
I6	Articles published before December 2021.

Table 3
Exclusion criteria for elimination of research papers.

Exclusion criteria	Description
E1	Exclude short irrelevant papers
E2	Exclude papers without proper content
E3	Exclude deviated papers from the survey
E4	Exclude papers with incomplete data

4.2. Inclusion and exclusion criteria

The inclusion and exclusion criteria for the selection of research papers by searching through databases is done and the PRISMA analysis diagram has been formulated.

Tables 2 and 3 displays the criterion for inclusion and exclusion of research papers.

Fig. 4 shows the selection of research data based on inclusion and exclusion criteria in five different phases.

PRISMA is an evidence-based minimum reporting set of items for systematic reviews and meta-analyses.

The advantages of PRISMA framework:

- Makes it possible for readers to assess the techniques strengths and weaknesses.
- Reviews are of high quality, structured and formatted.

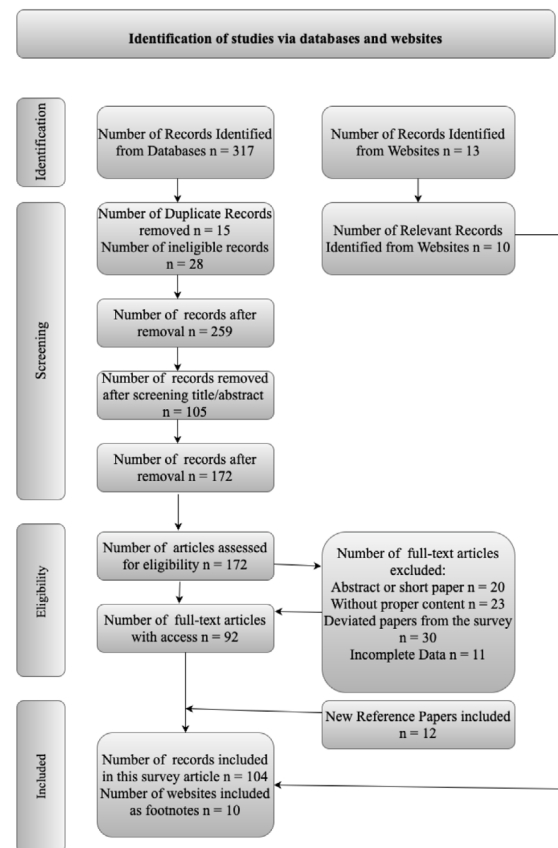


Fig. 5. PRISMA analysis diagram for the selection of papers.

- Permits replication of review methods.

Fig. 5 depicts the PRISMA analysis diagram by identifying, screening, and assessing the eligibility of research papers. Initially 317 research papers were collected and after the analysis and screening the total number of research papers has been refined to 104. The word cloud depicts the visual representation of most frequent words occurring in the selected articles as shown in Fig. 6. From the word cloud, it is clear that the words such as machine learning, data, patient, system, health care, application, research are most frequently repeated. These words gain a lots of importance in the context.

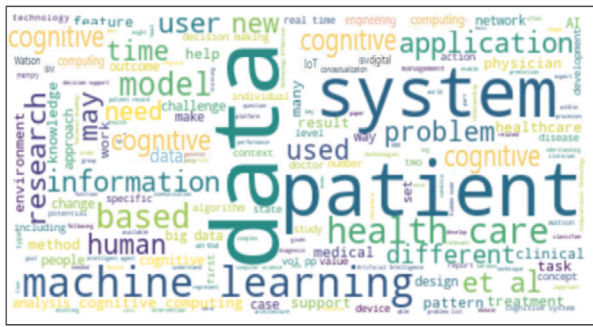


Fig. 6. Visual depiction of keywords in the surveyed research articles.

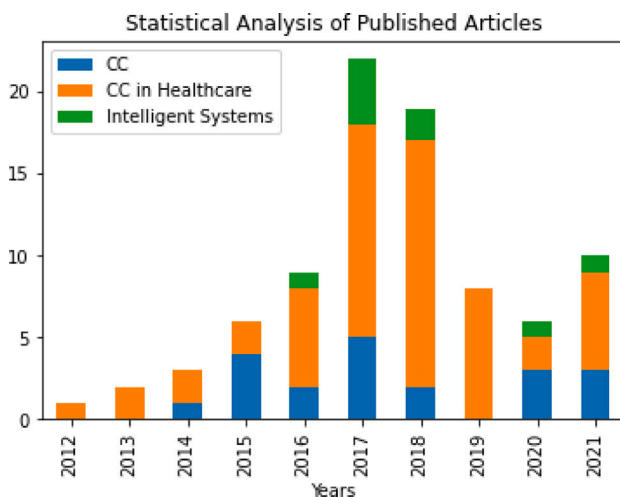


Fig. 7. Statistical analysis of published articles CC - Cognitive Computing.

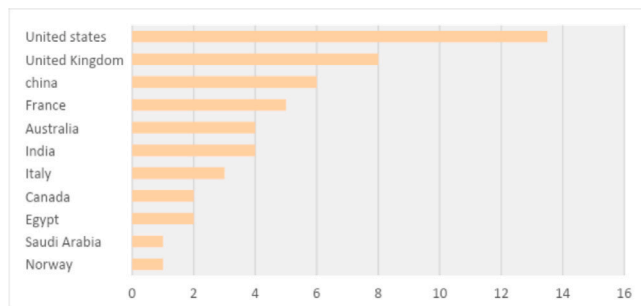


Fig. 8. Distribution of articles by country.

4.3. Statistical analysis of collected research articles

The collected research articles during the period from 2014 to 2021 are analyzed and the stacked bar graph is shown below. The Fig. 7 depicts the various domains of cognitive computing in healthcare. The analysis shows that in the year 2017 and 2018 many research papers are being published in the domain cognitive computing in healthcare. Nearly 13 publications are published in the year 2017 and 15 publications are published in the year 2018 indicating a huge demand of cognitive computing in healthcare. Fig. 8 depicts the distribution of articles country-wise. It depicts that the country United States has contributed the most articles on cognitive computing in healthcare compared to other countries.

4.3.1. Analytical synthesis

Analytical synthesis critically examines the gathered publications as (i) Study source by Country, Indexer and H Index, (ii) % of international collaboration, (iii) Distribution of studies by year and digital database, (iv) Count of source of studies by year, and (v) Detail list of papers considered for this study including citations. Table 4 indicates the study source by country, indexer and H-Index. From the Table, it is clear that the IEEE Access in country New York has high H-index of 158. Table 5 depicts the percentage of international collaboration. IEEE Access scores higher percentage from 2017 to 2021.

Table 6 explores the distribution of studies by year and digital database such as SCOPUS, IEEE Xplore, MDPI, Springer and ScienceDirect. The study includes high number of papers from SCOPUS database.

Table 7 describes the number of source of studies yearwise. Table 8 explores the detailed list of papers considered for this study including citations.

4.4. Research questions

The cognitive computing in healthcare raises the following research questions

- RQ1: How the cognitive computing domain addresses the challenges in healthcare?

The medical device reports, clinical trial results, patient medical records, and billing records, [43] are created and managed by the healthcare environment. In some cases, healthcare data are well-integrated, but in most cases, they are not. Therefore, analysis of the vast amount of generated healthcare data poses a great deal of challenge. Therefore, cognitive computing domain is needed to properly integrate the medical data on a large scale in a cost-effective manner.

- RQ2: What are the cognitive computing systems prominent in healthcare?

Watson For Oncology [44], Medical Sieve [45], NLP based cognitive system [46], cognitive Decision support system [47] and Watson patient record summary model [48] are the cognitive computing systems prominent in healthcare.

- RQ3: What are the future research directions of cognitive computing in healthcare?

The future research directions of cognitive computing in healthcare involve personalized medicine, knowledge and evidence-based medicine, better disease management, optimal treatment, disease outbreak prevention [36], precision medicine, diseases predictive modeling and organization of medical data.

- RQ4: What are the mathematical preliminaries required to implement cognitive computing applications?

The general knowledge on mathematical foundations [49] is necessary for performing intelligent computations. Cognitive Knowledge Base (CKB), hypothesis generation and evidence extraction, and inference generation are the mathematical preliminaries required to implement cognitive computing applications.

5. Synthesis of related work

In general, the three era's of computing [50] are tabulating systems (1900), programmable systems (1950) and cognitive computing systems (2011). The cognitive computing system manages huge volumes of structured, unstructured data and effectively performs tasks by applying self-learning algorithms. These systems also make use of NLP to extract meaning, draw inferences from large unstructured data, and interact with humans naturally. The three capabilities, namely understanding, reasoning, and learning, differentiate the cognitive systems from the traditional programmable systems.

A cognitive computing architecture called BriCA (Brain-inspired Computing Architecture) [41] has been designed by combining the

Table 4
Study source by Country, Indexer and H-Index.

Study source and study	Country	H-Index
IEEE Access (R1)	New York	158
IEEE Pulse (R2)	United States	89
The Journal of Arthroplasty (R3)	–	143
Journal of e-Learning and Knowledge Society (R4)	Italy	19
Future Generation Computer Systems (R5, R7)	Netherlands	134
Big Data and Cognitive Computing (R6)	Switzerland	18
International Journal of Engineering & Technology (R8)	Singapore	10
Sustainable Cities and Society (R9)	Netherlands	82
Health Information Science and Systems (R10)	Portugal	21
Procedia Computer Science (R11)	Netherlands	92
IEEE Sensors Journal (R12)	United States	132

Table 5
% of International collaboration.

Study source & study	2017	2018	2019	2020	2021
R1	38.25%	39.5%	24.8%	22.5%	45.87%
R2	3.8%	4.9%	2.9%	6.7%	8.4%
R3	–	–	–	–	–
R4	8.57%	9.43%	5.3%	6.4%	9.56%
R5, R7	32.33%	35.12%	20.19%	21%	29.18%
R6	21.2%	22.5%	13.5%	12.5%	21.5%
R8	–	–	–	–	–
R9	13.75%	20.5%	10.54%	8.64%	16.84%
R10	–	–	–	–	–
R11	20.5%	21.4%	16.2%	15.5%	22.5%
R12	23.16%	24.5%	18.2%	16.5%	24.6%

Table 6
Distribution of studies by year and digital database.

Digital database	Study	Year	No. of Papers
SCOPUS	R3, R4, R5, R8, R12	2018	5
	R2	2017	1
IEEE Xplore	R1		1
MDPI	R6	2017	1
Springer	R10	2017	1
ScienceDirect	R7	2018	1
	R9	2021	1
	R11	2015	1

modules of ML with the object language called JavaScript Object Notation (JSON). It is an ideal software platform for building and testing the hypothesis to exhibit human-level cognitive capabilities. In healthcare, [51] proposes a generic architecture called AI-based Healthcare system which automatically predicts and generates the results from its own experience. This system incorporates NLP, knowledge representation, automated reasoning, and ML.

Patient-centered cognitive support [52] emerged as a research challenge which enables the clinicians to picture a conceptual model called the virtual patient model. The clinician interacts with the model by sifting through the raw data, synthesizing the medical knowledge, and mapping the medical logic from the knowledge to provide cognitive decision support. The challenges in delivering patient-centered cognitive support are modeling, automation, data sharing, data collaboration, data management at scale, and full automated capture of doctor-patient interaction. The amount of data in EMR becomes a big question before the medical professionals. So, an EMR summarization model [48] has been developed to handle each patient record. This

Table 7
Count of source of studies by year.

Study source & study	2015	2017	2018	2019	2021	Total
R1				1		1
R2		1				1
R3			1			1
R4			1			1
R5, R7			2			2
R6		1				1
R8			1			1
R9					1	1
R10		1				1
R11	1					1
R12			1			1
Grand total	1	3	6	1	1	12

model organizes the medical data meaningfully by generating and relating the medical problems list to the clinical data.

An automated cognitive clinical health assessment using the activity behavior approach [53] has been devised to monitor the daily behavior of the people at home and predicts the clinical scores for each person. The Clinical Assessment using Activity Behavior (CAAB) approach models the behavior of residents and detects the clinical assessment score. This approach handles the two years of sensor data collected from 18 smart homes to generate the cognitive clinical score. The clinical assessment score is calculated as shown in Eq. (2)

$$S_1, S_2, S_3, \dots, S_p (p = T/m) \quad (2)$$

where $S_1, S_2, \dots, S_p$ represents the assessment score, $m = 180$ days which indicates the clinical tests, and T represents the time period. In future, the CAAB approach can be validated on larger population samples and clinical utilization of smart home based predictions must be explored.

For a novel medical care system [36] proposes a fifth-generation cognitive architecture. This system consists of a resource and data cognitive engine which assimilates speech emotion recognition capability. The cognitive engine identifies the physical status of victims and provides remote treatment for physiological diseases. The strength is the cognitive engine provides remote treatment for physiological diseases based on the victims' physical status. The major challenge lies in providing security to the patient's sensitive medical data.

Table 8
Detail list of papers considered for this study including citations.

Study	Reference	Country	No. of Citations	2019	2021	Total
R1	Amin et al. [32]	Saudi Arabia	144	1		1
R2	Ahmed et al. [13]	United States	93			1
R3	Bini [33]	United States	370			1
R4	Coccoli and Maresca [34]	Italy	12			1
R5	Chen et al. [35]	China	233			2
R6	[36]	China	151			1
R7	Gad et al. [37]	Egypt	57			1
R8	Jayanthi et al. [38]	India	–		1	1
R9	[39]	Pakistan	84			1
R10	Marshall et al. [40]	USA	21			1
R11	[41]	Japan	7			1
R12	[42]	Norway	64	1	1	12

An edge cognitive computing based smart healthcare system is a real-time healthcare system [35] has been proposed to mainly help the people in emergency situations to provide personalized patient care. This system continuously monitors the patients and assesses the risk of the patients and improves the survival rate. Two cognitive engines has been proposed such as data and resource. ML, DL and cognitive computing algorithms are employed in data engine. Technologies such as network softwarization and resource management are employed in resource engine. The strength is better user Quality of Experience was offered. The Quality of Service (QoS) quantization is performed to evaluate the performance of healthcare systems as shown in Eq. (3)

$$Overall\ QoS = \sum_{i=1}^m \frac{s(i)}{connC(\sigma(i))} \quad (3)$$

where $s(i) \in \{0, 1, 2, 3\}$ represents the health risk level of i th user, $\sigma(i) \in [1, n]$ indicates the edge node at present connected to user i , $connC(j)$, $j \in [1, n]$ indicates the total number of users connected to the edge node j .

Table 9 indicates the major role of cognitive computing in healthcare by presenting the article summary table with methods, algorithms, applications, results and limitations. In future, the emotional recognition system can be integrated with the system and personalized care can be provided.

5.1. Gaps in the existing literature and future trends

In the area of cognitive computing in healthcare, there have been only a few literature reviews, but none have provided crucial insights [43]. Consequently, it becomes more difficult to understand and apply the benefits of cognitive computing to improving patient care. It may also be challenging for researchers to track and use its capabilities and impact on healthcare. Few limitations of cognitive systems include limited risk analysis, meticulous training process and automated critical decision. These systems fail to analyze the data risks. The training process is meticulous because the entire training data must be understood by the system and then trained. Cognitive systems depend on humans for taking wise decisions so they lack in automated critical decision making.

Based on the related works presented above, the future trends of cognitive computing systems involve research optimization, personalized medicine, knowledge and evidence-based medicine, better disease management, optimal treatment, emotional intelligence, disease outbreak prevention [36], precision medicine, predictive modeling and

organization of medical data. In recent years, cognitive IoT has become increasingly popular [27,54]. The cognitive IoT sensor of the human body is used to detect body signals in real-time and can support real-time diagnosis and treatment services for clinicians.

6. Cognitive computing platforms

The various cognitive computing platforms in the field of healthcare such as

- IBM Watson
- Microsoft Cortana Intelligence Suite
- Google Deepmind
- SparkCognition's SparkPredict
- Expert System's Cogito
- Numenta's NuPIC
- CognitiveScale's Cognitive Cloud
- HealthTap's Dr. AI
- Enterra solution's Aila
- Healthcare X.O's Cognitive Assist

The mind map, as shown in Fig. 9, contains a summary of the cognitive computing companies indicated as amber-colored boxes. Brown colored boxes indicate the various cognitive computing products. Gray colored boxes indicate the various features of cognitive products. These platforms allow doctors, nurses, and medical professionals to view the real-time datasets for making evidence-based decisions.

6.1. IBM's Watson

IBM has put immense effort into cognitive computing, mainly in the field of healthcare for cancer treatments, clinical trial matching, personalized treatments, genomics, medical imaging, drug discovery, and so on. IBM progressed the research problem of open domain question answering through Jeopardy game show. Due to the success in the Jeopardy game show, IBM developed an AI platform for a business called Watson. Watson sifts through huge amounts of data [55] stored in servers and finds out the relevant meaningful and useful information. Watson understands and learns natural language, formulates the hypothesis and generates optimal outcomes

Table 9
Role of cognitive computing in healthcare.

Author, Year and Method	Algorithm	Application	Result	Limitation
[32] Cognitive healthcare model for pathology detection that fuses IoT-cloud technologies	VGG, AlexNet	Sensors to acquire multimodal medical data	86.59% & 87.32% of accuracy for VGG, AlexNet	Yet this model attains low accuracy rate.
[13] How IBM Watson platform improves and ensures ability by using massive amounts of health data.	Hypothesis generation, dynamic learning, NLP	Data about patients is collected in large volumes, including EHR, genomic and exogenous data.	Real-time analysis of health information to make informed decision	An in-depth training process is necessary for risk analysis
[33] How AI support and enhance human cognitive functions for doctors	ML, ANN, IBM's Watson	Physicians provide intensive care for critical patients	Improves the accuracy of diagnostics	Regeneration and maintenance are incredibly costly.
[34] Analyses the use of cognitive computing-based healthcare	Enterra, DL, IBM Watson, DeepMind, IDOL	Medicine healing and visual pattern recognition	Cognitive computing Machines reason as humans in healthcare	Inadequate infrastructure and high hardware requirements
[35] Edge-Cognitive-Computing-based (ECC) smart- healthcare system to monitor and analyze health	ECC-based smart healthcare system- ML, DL, cognitive computing	A patient's emergency conditions	Significantly increases patient survival rates in sudden emergencies	Emotions are not utilized in the suggested technique
[36] Fifth-generation cognitive architecture	Resource and data cognitive engine	Speech emotion recognition capability	Recognizes the emotion of users' speech.	Securely storing sensitive medical data is major challenge
[37] IoT Iris Recognition using Multi-Algorithmic techniques	Delta-mean multi-algorithm-mean	Iris Recognition	99.4% accuracy	Fuse scores in identification mode are missing
[38] Cognitive computing and personalized medicine	Cognitive computing, IBM Watson	Personalized Medicine	Analysis between traditional and personalized medicine	Massive infrastructure investment is needed for personal medicine
[39] Cognitive Assessment of Smart Home Residents (CA-SHR) to evaluate residents' abilities using pre-determined scores	Ensemble AdaBoost	Pre-segmented activities of training in smart-homes	96.02% and 99.6% of accuracy	Suggested model cannot assess human activities in real-time
[40] A computer-based human problem-solving framework AI-based research models and Big Data as a catalyst. and forecast	Deep ML	Health and life science approach	Offers a prospective look at how AI will affect human-computer support in health and life sciences.	A lot of raw data is required by systems to gain insight, learn,
[41] Cognitive computing, ML	Brain-inspired Computing Architecture	Brains enable us to build intelligent machines with human-level cognitive abilities	Version 1 of BriCA is licensed under Apache License and implemented in Python	Inter-neuron processing is slower than maximum firing rate of 1kHz
[42] Body sensors and deep CNN, an activity recognition method for smart healthcare	Gaussian kernel based PCA and Deep CNN	A person with body sensor for medical monitoring	93.90% accuracy	No cloud-based platform to create smart home environment

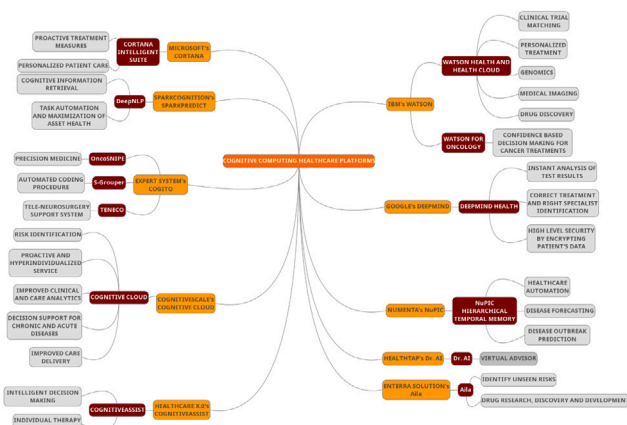


Fig. 9. Cognitive computing companies and products.

6.1.1. Working of IBM Watson

Watson is a question answering system capable of answering the natural language questions. IBM Watson fuses the techniques of AI, ML, NLP to design logic and data based natural language insights [28]. IBM Watson works with three major processing engines such as NLP, big data analytics, and knowledge reasoning [56]. Watson analyses millions of documents by making use of thousands of rules to derive the best and optimal answer.

IBM Watson follows a sequence of steps [17] namely natural language input analysis, question decomposition, search analysis, hypothesis generation, evidence extraction and confidence score determination. Watson takes natural language as input and analyzes the question with the help of 100 algorithms or more to determine the question type. The question is then divided into sub-questions and each sub-question is examined individually. In the search analysis and hypothesis generation step, the system searches answers to question by analyzing the knowledge base using hundreds of thousands of logic rules. Many search techniques namely passage, single entity can be used. In the

next step, the hypothesis are scored using scoring algorithms and their evidence is extracted. Evidence less hypothesis is refuted and evidence based hypothesis is accepted. Watson then ranks the hypothesis and renders confidence values along with probability to ensure that the answer is correct. Thus IBM Watson helps in evidence based decision making, boosts quality of care by rendering cost effective systems.

IBM's cognitive health [56] products development is increasing rapidly. In April 2015, IBM developed the Watson Health and Watson Health Cloud [57] to improve the ability of physicians and researchers. Watson Health and Watson Health Cloud delivers next generation medical solutions. IBM Watson Health² obtains reasonable insights, delivers customized and optimized treatment plans from massive amounts of health data. Watson integrates nearly 23 million medical papers from the National Library of Medicine (MEDLINE), and it can access that data in milliseconds.

Watson is applied in healthcare domain based on DeepQA technology [58] and renders evidence based decisions. This article describes the working of DeepQA architecture in the healthcare domain. The DeepQA architecture [59] is an efficient and powerful architecture for reasoning and analyzing the natural language. The main goal of using DeepQA in healthcare is to minimize the input information, automate the optimal extraction of medical records and generate evidence based hypothesis and decisions. Several challenges addressed in this paper are extraction of useful and meaningful clinical information, identifying the missing information about the patients, framing hypothesis and verification components and performance analysis. The system is evaluated on 188 doctor's uncertain diagnosis questions.

Watson health [6] reads and analyses many millions of information in just few seconds. IBM Watson health employs many different cognitive technologies [4] for data analysis and discover new inferences and insights from the existing information. By this way, IBM Watson health provides personalized treatment. This healthcare system assists the physicians in detecting the various diseases at an early stage more precisely and accurately. Watson health has been used by Medtronic to develop glucose monitor and insulin pump for diabetes by analyzing nearly 125 million patient details. This system automatically monitors the levels and supplies sufficient amount of insulin.

Watson computing system processes huge amounts of data from thousands of sources. Some of the key features of Watson [60] are natural language processing analytics for unstructured data, data-driven decision support, patient similarity analytics, metagenomics, evidence-based case structuring (evicase), and streaming analytics. The major component to be focused is the evicase framework. The three phases of evicase framework are knowledge encapsulation, retrospective analysis, and prospective analysis. Knowledge encapsulation deals with developing rule sets and ontologies and provides the patient's guideline view. Retrospective analysis generates useful insights about the patients, follows guidelines, and assesses outcomes. Prospective analysis deals with the design and development of statistical ML models to render evidence based treatment recommendations and improved outcomes. The drawbacks of Watson system are requirement of huge amounts of data, uncertainty or inconsistency of data. In future, the evidence based decision support feature of Watson must be analyzed to deliver an efficient system.

Watson augments the doctor's capability by providing the best information from the incredibly huge amount of information. The cognitive technology plays a major and important role in cancer. Mainly in the field of oncology, the systems assist in developing an individual treatment plan for patients. Watson serves as a supercomputer [61] to take timely informed and optimal decisions. IBM Watson generates optimal answers using statistical analysis and accumulated information. Watson follows a continuous learning process by analyzing question, extracting hypothesis and generating evidence. Watson becomes smarter by

getting continuous feedback from the users. Cancer treatments are complex, so these cognitive technologies are necessary for deriving evidence based treatments.

IBM Watson for Oncology [62] has been designed to help the oncologists in confident decision making for cancer treatments. This system derives data from various medical journals, textbooks, and million pages of text. In IBM Watson for Oncology (WFO) [28], patient's medical queries are given as input and a set of hypothesis is listed with multiple treatment options. This treatment options helps the physicians in right decision making at the right time. A clinician uses Watson [34] which takes only 20 s to collect and process patient's information. So, the speed of Watson's diagnosing capacity is very fast, precise and accurate. Using Watson, the processing time is reduced and mistakes are avoided. IBM WFO [63] has been used by physicians to provide evidence based treatment for patients with cervical cancer.

Application of cognitive computing technology in WFO [44] has a great potential in order to improve the quality of treatment. WFO systems serves as evidence based clinical decision support computing system for providing personalized patient care. This WFO not only helps the medical professionals by rendering green colored first choice recommendations but also helps the people who are not familiar with the disease by rendering yellow color recommendations. As the amount of patient's data increases, better and optimal decisions of WFO can be obtained. The concordance rate is calculated between the WFO treatment options and the decisions of oncologists as shown in Eq. (4)

$$CR = \frac{\text{No. of concordant subjects}}{\text{Total No. of subjects assessed}} \quad (4)$$

WFO [62] is an AI-based clinical decision support system designed for the treatment of breast cancer. WFO gives full support to the oncologists for making optimal treatment decisions by sifting through a knowledge base containing many journals, articles, textbooks, etc. It is a powerful cognitive technology platform that makes use of NLP and ML techniques for analyzing the patient's health, generating inferences, and providing personalized treatment. The concordance rate is calculated as shown in Eq. (4) between tumor board suggestions and WFO list of suggestions. The increase in concordance rate indicates that the WFO list of suggestions or recommendations highly resembles with the oncologists treatment suggestions.

Another interesting application of IBM Watson is Sugar.IQ [64]. Sugar.IQ application delivers timely and personalized insights to the users and their feedbacks are analyzed by fusing AI, ML and analytics technologies. When Watson is provided with a particular patient's medical report [33], it will suggest the most likely treatment to be followed by considering the patient's medical history. Watson when ingested with more information produces more accurate and precise results. These technologies would make doctors more efficient by augmenting human intelligence and helps them in evidence based confidence weighted decision making. The strength is that Hospitals will employ analytics based on historical data to streamline workflows and resource allocation.

Watson for Genomics (WfG) has been utilized [65] for providing Next-Generation Sequencing (NGS) guided cancer care. After sequence completion, the information such as tumor type, list of variants, gene level copy number has been uploaded to WfG. The process involved in WfG are molecular profile analysis, actionable alterations analysis, drug analysis, and report generation. WfG has been infused with NLP to extract reasonable and actionable insights from the unstructured data of the patients who are involved in clinical sequencing program. WfG ingests massive amounts of data such as thousands of cancer related publications and renders useful insights as rendered by human only Molecular Tumor Board (MTB). The pathway to NGS based cancer care is an iterative process for providing actionable and targetable insights from the patient's reports. WfG uses NLP techniques [66] to extract automated evidence based decisions for breast cancer patients. NGS was used to obtain clinical trial and treatment options and delivers new information to the physicians.

² <https://www.ibm.com/watson/health/>

To summarize, IBM Watson ingests all kinds of data and acts as an evidence based decision system which reasons out and analyses massive volumes of patient data. IBM Watson also provides solutions for medical image analysis, disease progression and management, deep data analysis, and evidence based treatment.

6.1.2. Microsoft's Cortana

Microsoft's Cortana Intelligence Suite³ is an example of an AI platform which turns data into intelligent action. Using the previous history of patient's data from hospital visits, hospitalized period, lab records, Electronic Medical Records (EMR), Electronic Health Records (EHR), admission records, etc. Cortana intelligence suite builds models that can learn from the previous history, extract patterns, and provides proactive treatment measures. This model also offers personalized patient care.

6.1.3. Google's Deepmind

Google's Deepmind⁴ is cognitive computing software for solving intelligent real-world applications. The aim is to solve problem with intelligence [34] using AI technology that can learn by itself. In 2015, Google created Deepmind health enables the doctors to analyze the test results instantaneously, identifies the correct treatment, and accelerates the special cases to the right specialist. It provides the highest level of security by highly encrypting the patient's data. It makes use of deep reinforcement learning technology to have human-level control. This software renders different open source platforms, datasets and codes to solve complex problems [4].

6.1.4. SparkCognition's SparkPredict

SparkCognition⁵ is another cognitive computing software which provides SparkPredict, SparkSecure, and MindFabric to boost the human intelligence at machine scale. It aims at revolutionizing the information into litigable perceptions that help the people to precognit the future and make informed decisions. DeepNLP produced by SparkCognition deals with cognitive information retrieval. Cognitive Fingerprinting algorithm builds the model which automates the task and maximizes the asset health.

6.1.5. Expert System's Cogito

Expert System's cognitive software called Cogito⁶ imitates the ability for reading, reasoning, and understanding. Cogito serves as an Application Programming Interface (API) for NLP and text analytics. This software fights against tumors manage patient data and reduce healthcare costs. This software is also engaged in the projects like OncoSNIPE (creates precision medicine for patients resistant to cancer treatments), S-Grouper (activates automatic coding procedure for hospital activities) and TENECO (tele-neurosurgery support system).

6.1.6. Numenta's NuPIC

Numenta inspired by machine learning technology based on the theory of neocortex (part of cerebral cortex involved in higher order brain functions such as sensory perception, conscious thought and so on). Numenta developed a project called Numenta Platform for Intelligent Computing (NuPIC). It is a machine intelligence platform which implements Hierarchical Temporal Memory (HTM). HTM deals with the theory of brain and a set of layouts for machine intelligence. NuPIC performs automation in healthcare by the prediction of diseases based on the patient's symptoms. A prototype application called MATE [67] is built based on Numenta's HTM for disease forecasting and disease outbreak prediction.

6.1.7. CognitiveScale's cognitive cloud

CognitiveScale provides cognitive cloud software⁷ for cognitive computing, which is a new class of big data analytics and ML systems. Cognitive cloud software services in healthcare include improved care delivery, patient and member engagement, pattern detectors to identify risks, improved clinical and care analytics, and decision support for chronic and acute diseases.

6.1.8. HealthTap's Dr. AI

HealthTap's Dr. AI⁸ is a virtual advisor developed as a result of a cognitive solution. It entrusts on the medical profiles of the patients, and the intelligence gathered from nearly 105,000 doctors. It accumulates the list of symptoms and connects to the doctors by incorporating both context and clinical expertise of doctors.

6.1.9. Enterra Solution's Aila

Enterra Solutions⁹ developed an enterprise cognitive system called Aila. This system provides an AI platform by combining cognitive computing with NLP, big data to obtain data-driven insights from the unstructured clinical data. This system focuses on drug research, discovery, and development to identify unseen risks, cares, and treatments.

6.1.10. Healthcare X.O's CognitiveAssist

Healthcare X.O generated software called HealthAssist, which creates the patient data into a digital file and explains the progress of treatment. This software consist of the following modules: VascularAssist, CardioAssist, TeleAssist, WoundAssist, DiabetAssist, EndoAssist, CognitiveAssist, OncoAssist, and PatientAssist. CognitiveAssist¹⁰ helps the medical professionals in intelligent decision making by combining terabytes of patient data and mapping it with the medical journals, research results, and clinical studies. It also helps in creating an individual therapy plan for each patient.

This section presents various cognitive computing platforms and products with their distinguished features. From the mind map, it is clear that IBM Watson is an effective cognitive assistant helpful for doctors because IBM Watson offers more services and applications such as question answering system, IBM's cognitive health, Watson Health Cloud, healthcare domain based on DeepQA technology, IBM Watson for Oncology, Sugar.IQ, and Watson for Genomics (WfG). IBM Watson is an AI-based supercomputer that functions as a "questions-answering" computer using sophisticated analytical software. It is believed that Watson is capable of solving life-threatening problems like cancer, oncology, etc. The IBM Watson performs analytics on vast data repositories to answer human questions quickly.

A line graph, as shown in Fig. 10, portrays the yearwise development of various cognitive computing platforms. These products provide unique benefits to people by augmenting their intelligence level. From the graph, it is clear that IBM Watson is the first cognitive computing platform developed in the year 2011. In the year 2014, nearly three cognitive computing platforms, namely Cortana, Deepmind, and SparkPredict, were developed. Enterra solution's Aila is the recently developed platform helpful for the computers to think and act on their own by mimicking human's mind.

The segmentation of the cognitive computing market is based on businesses function. Solutions for cognitive computing can help businesses become more cost-effective and efficient in their operations. Many businesses are implementing cognitive computing solutions and services to improve operations and achieve the highest levels of employee engagement.

³ <https://azure.microsoft.com/en-in/try/cognitive-services/>

⁴ <https://deepmind.com/>

⁵ <https://www.sparkcognition.com/>

⁶ <http://www.expertsystem.com/products/cogito-cognitive-technology/>

⁷ <https://www.cognitivescale.com/solutions/healthcare/>

⁸ <https://medium.com/@HealthTap/dr-a-i-80b4cf06be30>

⁹ <https://www.enterrasolutions.com/life-science/>

¹⁰ <http://www.healthcare-xnull.com/>

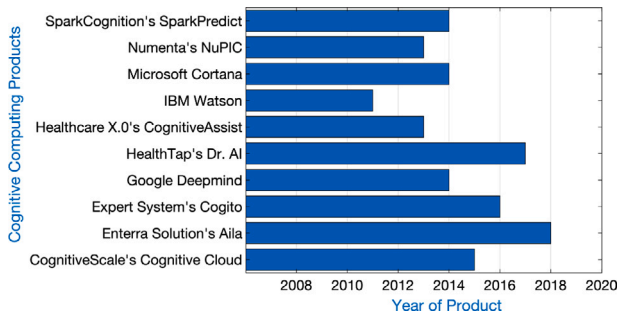


Fig. 10. Yearwise development of various cognitive computing platforms.

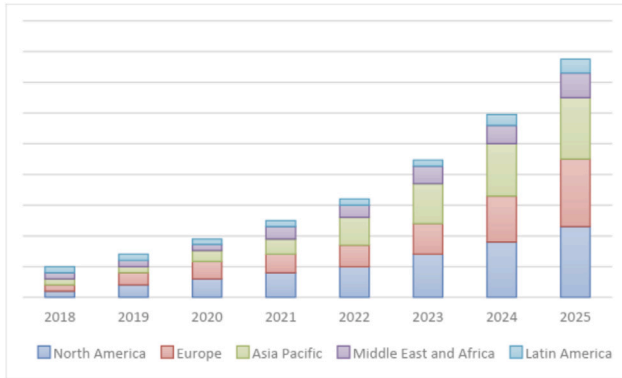


Fig. 11. Market capture of cognitive computing products.

Fig. 11 illustrates the market capture of Cognitive computing products. According to the plot, North America is predicted to hold the largest share of the cognition computing market, while Asia Pacific is predicted to grow the fastest.

7. Cognitive tools and frameworks

Cognitive computing technology augments human decision making and develops brain-like computers by imposing human intelligence. Certain cognitive tools and frameworks are required to scale up the clinician's expertise. The libraries, tools and frameworks used for implementing cognitive applications [5] are TensorFlow, Apache Singa, Torch7, Theano, Caffe, Pytorch, Massive Online Analysis (MOA), MLlib, Pattern, Scikit-learn, Shogun, Veles, Deeplearning4J, neon, H2O, Mahout, Apache R, Amazon ML, Azure ML Studio.

A cognitive framework such as Watson NLP pipeline engine analyses the unstructured data and makes it useful to the medical teams for further diagnosis and render the right treatment for the patients. The pipeline engine performs sentence segmentation, tokenization, Parts Of Speech (POS) tagging, parsing, medical concept annotation, relation detection, medication extraction, lab and vitals extraction, note section classification and goal extraction. Most commonly used open source tools and frameworks [4] for performing NLP operations are Apache Lucene Core, Apache Unstructured Information Management applications (UIMA), openNLP, Open Text Summarizer, Stanford NLP, SyntaxNet and so on.

Another interesting application for performing deep learning operations in cognitive platform is the Microsoft Cognitive Toolkit (CNTK) [34]. CNTK explores neural networks using directed graph in distributed environment. CNTK also implements the hybrid models of deep neural networks such as Recurrent Neural Network (RNN), Convolutional Neural Network (CNN) and Long Short Term Memory (LSTM). The strength is that health care will be impacted greatly by cognitive health care.

A ML technique has been proposed in IBM Watson [68] to automatically derive the patient's medical details. The system makes use of IBM Watson NLP pipeline to extract the lexical and medical features from the patient's medical records. This system contributes to an accurate generation of medical problem lists. It has been implemented as a pilot study which makes use of IBM Watson methodology which randomly chose 15 patient records and they were reviewed by two medical experts. The experts started comparing their problem list (P) with the Watson problem list (W) and EHR problem list (E) using 10 point scale. Some of the major features of W is frequent problems, term frequency, medication score diagnosis, and probability. The major issue of this system is inaccurate patient information and less problem list maintenance. In future, the EHR problem list can be checked and analyzed to include more important symptoms of the disease.

7.1. Mathematical preliminaries required for cognitive computing applications

The general knowledge on mathematical foundations [49] is necessary for performing intelligent computations. Cognitive computing in healthcare scales, augments, accelerates the expertise of medical professionals and equips them to handle next-generation healthcare data effectively using cognitive technologies. Cognitive computing also offers an evidence based clinical decision system. The Cognitive Knowledge Base (CKB) is closely related to healthcare, in which the defined mathematical preliminaries are applicable in healthcare. Moreover, the AI-based healthcare system creates a patient map based on health condition for personalized patient care by making use of NLP and ML algorithms. This system is helpful in real-time monitoring of patient's vitals by using wireless sensors. The mathematical preliminaries required for CKB, hypothesis generation and evidence extraction, and inference generation is directly related to the cognitive computing process. Artificial Intelligence techniques in combination with computational theory renders cognitive computing methods. Probability and statistics, concept analysis, concept algebra, semantic analysis, semantic algebra, relation algebra, hypothesis testing concepts are essential for implementing cognitive applications.

Denotational mathematics represents the high level mathematical structures such as objects, relations, concepts, behavioral information, process and knowledge. The different paradigms of denotational mathematics are concept algebra, system algebra, real time process algebra, and visual semantic algebra. Denotational mathematics is applied rigorously to formal concepts in concept algebra. Concept algebra is used in cognitive computing, cognitive knowledge bases, cognitive translation, and cognitive search engines.

The concept algebra [69] can be defined as shown in Eq. (5)

$$CA \triangleq (C, OP, \theta_C) = (O, A, R^C, R^I, R^O, \cdot, \cdot, \cdot, \theta_C) \quad (5)$$

where O represents objects, A represents attributes, R^C represents set of internal relations, R^I indicates set of input relations, and R^O indicates set of output relations.

7.1.1. Cognitive knowledge base (CKB)

A CKB is an intelligent knowledge base which discovers the hidden semantic information from huge amounts of unstructured data. Cognitive knowledge bases CKBs are novel structures of intelligent knowledge bases that mimic human knowledge processing by representing and manipulating knowledge as dynamic concept networks. CKBs are based on mathematical models of formal concepts interconnected across an ontology or rule-based knowledge base by dynamically associating formal concepts with each other. Concept algebra-based algorithms for knowledge acquisition and retrieval control the CKB system. For cognitive robots and machine learning, CKB acts as the brain of the cognitive learning engines. The design of CKB has been proposed in [70] to generate Object Attribute Relation (OAR) model using Formal Concept Analysis (FCA). The CKB construction deals

with topic-model based concept extraction, determination of relevant objects, relevant attributes, hyper-attributes and generation of relational hierarchy. The concepts are extracted based on the Hierarchical Dirichlet Process (HDP) algorithm and Concept Scoring Function (CSF). The CSF is depicted as shown in Eq. (6).

$$CSF = \sum_{documents} TP \times TF of \text{ Noun Phrase} \quad (6)$$

where TP is the Topic Probability and TF is the Term Frequency.

To determine the relevant objects and attributes, For each object and attribute, the average information weight is calculated as shown in Eq. (7)

$$I_i = P(x_i / y_i) \times \log_2 P(x_i / y_i) \quad (7)$$

where x_i — $Object_i$, y_i — $Attribute_i$, I_i — Average Information Weight, $P(x_i / y_i)$ is the conditional probability.

The total average information weight is shown in Eq. (8)

$$I = \sum_{i=1}^m I_i \quad (8)$$

The hyper-attributes are determined as shown in Eq. (9)

$$Hyper-attributes = R_O \times R_A \quad (9)$$

where R_O indicates relevant objects and R_A indicates relevant attributes.

The relational hierarchy for the concept is constructed using the semantic knowledge algorithm.

Finally, the CKB is defined as shown in Eq. (10)

$$CKB = (O, A, R) \quad (10)$$

where O represents the set of objects $O = \{o_1, o_2, \dots, o_i\}$

A represents the set of attributes $A = \{a_1, a_2, \dots, a_i\}$

R represents the set of relations $R = \{r_1, r_2, \dots, r_i\}$

The major drawback is that the semantic analysis and algebra is not taken into account. In future, concept and semantic algebra analysis and operations must be incorporated.

7.1.2. Hypothesis generation and evidence extraction

Hypothesis generation starts with generating possible answers or candidate answers and evidence extraction step derives evidence for the hypothesis by generating confidence scores. The evidence based hypothesis are accepted and evidence less hypothesis are refuted. In future, the hypothesis testing mathematical concept can be applied for this process.

The t-test distribution can be applied as shown in Eq. (11)

$$t = \frac{(\bar{x} - \mu)}{(s / \sqrt{n})} \quad (11)$$

where $\bar{x}$ is the sample mean, μ represents population mean, n indicates sample size, s represents standard deviation.

For hypothesis testing, the doctors make use of Wilcoxon signed rank test [68] to compare the response scale ratings of Watson problem list and EHR problem list. This test is used to compare two similar or matched lists such as Watson problem and EHR problem and makes assessment on how the population means differ. This test is defined as shown in Eq. (12)

$$W = \sum_{i=1}^{N_r} [sgn(x_{2,i} - x_{1,i}) \cdot R_i] \quad (12)$$

where W is the Wilcoxon signed test statistic, N_r is the size of the samples not including $x_1 = x_2$, sgn represents the sign function, $x_{2,i}$ and $x_{1,i}$ indicates the Watson problem list and EHR problem list, R_i is the rank. The users rated that Watson problem list is much higher than EHR problem list. The p -value has been used for comparisons.

7.1.3. Inference generation

The WatsonPath system [71] analyses the scenario, constructs assertion graph, identifies the nodes of importance, generates relations, determines confidence in nodes, identifies hypothesis, and refines hypothesis based on evidence. WatsonPath system makes use of graph datastructure $G = \{V, E\}$ where V is the vertices and E is the edges to generate relations and probabilistic inferences. The graph datastructure employed is the assertion graph.

Finally, the Confidence Weighted Score (CWS) for the generated hypothesis is defined as shown in Eq. (13)

$$CWS = \frac{1}{n} \sum_{i=1}^n \frac{\text{number correct in first } i \text{ ranks}}{i} \quad (13)$$

where n is the number of questions. Thus, cognitive computing frameworks and mathematical preliminaries enhance human intelligence and help them in the construction of dynamic self-learning systems. These tools also build computational and decision-making models, which can reason, think, learn, and adapt to the healthcare environment.

8. Cognitive applications in healthcare

Cognitive computing technology augments the human's ability to understand and act upon complex volumes of data. Cognitive solutions in healthcare provide new capabilities to healthcare organizations. In other words, the goal of cognitive computing in healthcare [16] is not to replace the doctor but to augment the doctor's capabilities by processing the ginormous amount of information and providing evidence-based treatment. For example, to predict the risk of diseases such as diabetic retinopathy processing and analysis of medical images with deep learning, neural networks, and cognitive computing technology creates systems of intelligence.



Fig. 12. Applications of cognitive computing in healthcare. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The mind map pictured in Fig. 12, describes the various cognitive applications in healthcare (magenta colored boxes), cognitive assistants, and technologies (green colored boxes). The remaining ash-colored boxes illustrate the process of various cognitive assistants and technologies. These applications renders augmented treatment, informed decision making, personalized with recommended treatment options, automated self-help treatments and patient centered coaching by continuously monitoring the health status.

8.1. Cognitive applications

An interesting health application called Cognitive Concierge (cognitive in-patient concierge system) [72] deals with cognitive patient care. This application was designed to provide Cognitive Concierge Service

for hospitals by the combination of cognitive capabilities and Internet of Things (IoT). The design pattern comprises of Watson IoT platform (Bluemix), which consists of Watson Speech To Text (STT), Watson Conversation, Watson IoT storage, and Watson Text To Speech (TTS).

For example, Cognitive Concierge application warns patients with asthma or any other lung disease. Cafewell Concierge by Welltok is a cognitive powered tool that can instantaneously process huge volumes of data to answer the questions asked by the individuals and renders accurate recommendations to guide the people in achieving optimal health.

A cognitive smart IoT healthcare system [32] has been designed with deep learning Convolutional Neural Network (CNN) technique and IoT cloud technologies to continuously monitor the patient's state and render exact, timely results at low cost. The cognitive model infused in this system delivers evidence based real time decision. The strength is results are shared with healthcare professional they can offer emergency assistance if a patient is in a critical condition. An intelligent system called IBM Watson For Oncology (WFO) has been devised [73] for the prognosis and assessment of cervical cancer. It provides three therapy options such as recommended, for consideration, and not recommended. The main challenge in this system is to select the patients for clinical trial matching.

8.2. Hospital management

Cognitive computing brings a drastic change in the healthcare sector by setting up the Alder Hey Children's Hospital as the UK's first "Cognitive" hospital [74]. This cognitive hospital makes use of Watson, which provides meaningful insights from data by applying cognitive analytics. Nearly 275,000 children and young people are treated every year by this hospital. Thus, the Alder Hey Children's Hospital serves as a cognitive revolution for children's healthcare. Manipal Hospital is the first hospital in India to use Watson for augmenting the oncology treatment.

8.3. Clinical decision support and predictive analytics

Chronic diseases are the leading causes of death, and these diseases pose major challenges in healthcare. The increasing burden of chronic diseases drives the healthcare cognitive computing market. So, a Medical Cognitive System [2] has been developed for medical professionals to assist the doctors in the diagnosis and treatment of chronic diseases. This system fuses AI, ML, NLP to enable cognitive computing to make the right decision with various types of data and contents. Cognitive Decision Support System [47] for diagnoses the severity of the symptoms by knowledge base model and Fuzzy Cognitive Map (FCM). The mathematical formulation for the FCM is given in Eq. (14).

$$X(t) = f(W^T X(t-1)) \quad (14)$$

where X is the new state vector which contains values of n concepts, W indicates the weight matrix and t indicates the time. The system can also render collective opinion knowledge based on the key diagnosis of problems and severity of the disease.

A decision support system tool called Virtual Doctor System (VDS) [75] has been designed and developed to enhance doctor-patient relationships. This system can perform a complex and exhaustive analysis of data collected from various sources. The main feature of this system is to interact with the medical doctors to extract the knowledge base and construct a separate data mart for each data source. VDS serves as an assistant to the older people by prognosing and treating their diseases. CREDO schema develops cognitive systems at the point of care [76]. These cognitive systems support intellectual functions such as health status, risk measurement, decision making, treatment planning, progress management, and complete discerning of clinical proficiency.

A cognitive healthcare based support system has been designed [77] to analyze the health state of patients in order to prevent the disease. This system takes the sensor data, patient history of medical records, medical report, physical test and performs denoising, feature extraction, disease type categorization, and risk stratification. DNN is used to predict the disease. This system renders prior information and also determines the severity and progression of disease. The main challenges of this system are diagnosis with insufficient information, many sources of input, and creation of huge knowledge base.

Cognitive computing based clinical decision support system [78] has been developed for prediction of coronary cardiac disease. Two phases are involved where one phase deals with the fuzzy logic for heart disease identification and another phase deals with the application of DNN for early and automated prediction of diabetes disease. This cognitive model helps both the users as well as doctors by analyzing the patient's records and providing meaningful information. In future, the coordination of information by the management can help in more accurate diagnosis.

A clustered medical recommender system has been developed [14] categorizes patients and generates similar medical reports for patients with similar medical conditions. Two different tools such as IBM Watson and Framester is used for analysis. This system is very helpful for taking proactive actions and decisions. It makes use of IBM natural language understanding framework for understanding the patient's conditions.

A cognitive healthcare based system for building clinical decisions has been proposed [3]. Several case studies such as patient evacuation, self-management, healthcare concierge, and helped living are analyzed. Patient evacuation system deals with the design of dynamic scheduling system which analyses the emergency and critical situations of patients and renders them the treatments. The self-management system manages the patients with Type-II diabetes by developing a macro cognitive system for checking the sugar levels and also by using the knowledge base for managing the sugar levels. The healthcare concierge performs day by day analysis and gives information about the patient's health status. The helped living application has been developed for old people based on the knowledge base technology of cognitive system to help them.

8.4. Virtual assistants for patient monitoring

A new application for cognitive systems has been emerged using IoT based Consumer Electronics (CE). Microsoft Kinect [79] is applied in cognitive systems for introducing smart Consumer Electronics (CE). Certain projects based on cognitive assistants [27] have been developed by applying cognitive IoT in healthcare. DayGuide assistant has been developed to help the older people suffering from mild cognitive impairment.

An automated cognitive assistant called Medical Sieve [45] has been developed for the radiologists and cardiologists to help them in informed decision making. This assistant collects the large amounts of textual and imaging data of patients from EMR and filters only the essential information needed by the physicians for effective decision making.

An autonomic and cognitive IoT based system [54] has been designed and developed for managing and monitoring the patient's health based on their wearables. A set of autonomic cognitive patterns are derived and the decisions are provided. The system involves three processes such as process management, semantic management, and big data management. The system is evaluated using the response time. In future, some transformation rules are framed for refining the system and the optimal configurations are derived for enhancing the response time.

8.4.1. IoT in healthcare environment

In recent days the most challenging issue faced by healthcare is decision-making [80] when it comes to the installation of the right lightweight authentication cipher that can solve the authentication issues currently prevailing in IoHT devices. So, the suggested method selects the best lightweight cryptographic ciphers according to the most important parameters or criteria. Based on an evaluation of the 10 cryptographic algorithms, this framework selects the best lightweight cryptographic algorithm. In order to provide a fully pledged secure IoHT system, the security evaluation criteria cover all dimensions of lightweight cryptographic security.

Mobile healthcare brings massive support to IoT in the healthcare environment [81] through mobile computing. A systematic literature review protocol is used in this study to examine how mobile computing supports Internet of Things (IoT) applications in healthcare, contributes to ongoing and future research on IoT in the healthcare system, and influences IoT in the healthcare system.

Security is a major problem in the Internet of Health Things. Fragile patient data is transmitted from IoT devices to servers in healthcare environments based on IoMT. If there is any eavesdropping or intrusion during this transmission, data will be handled maliciously for malfeasance as well. In order to assess the security features of IoHT-based devices [82] in the healthcare environment, the Identified Security Attributes (ISA) framework is presented. Analytical Hierarchical Process and Technique for Order Preference by Similarity to Ideal Solution are two hybrid MCDM techniques used in the suggested framework. This framework operates in two stages: the first stage determines the weights of attributes using the AHP method, and the second stage performs a security assessment of alternatives based on security criteria using the TOPSIS method. The results of the suggested security assessment framework show that the IoMT system chose the most trustworthy and secure alternative among the alternatives.

A healthcare system built on early decision-making [83] has enormous potential to lower costs, improve quality, and lower waste and error. By aiding in the processing of enormous data volumes, complex system modeling, and sourcing derivations from healthcare data and simulations, scientific programming play a significant role in overcoming the current issues and future problems involved in the management of large scale data in healthcare. Consequently, in order to effectively address this issue, a thorough study and to assist doctors and practitioners in making decisions regarding the diagnosis of the disease and the appropriate course of treatment, analysis of the literature work currently in existence is necessary. The doctors and practitioners will be able to use this analysis of the existing research work to help them make more honest decisions, which will ultimately help them use the study as support for treating patients and making appropriate medicines recommendations.

In order to maintain smooth healthcare activities, [84] it is important to maintain security in the communication among devices. This limitation will be addressed by the suggested study's systematic literature review (SLR) for the IoHT security and provenance. This suggested framework put a focus on the IoHT's security analysis and provenance while also responding to the study's stated questions.

8.5. Cognitive technology for coaching patients

A customized cognitive coaching technology model [85] has been developed for personalizing healthcare and for the establishment of patient-centered infrastructure. This framework is a wearable, sensor-driven, and cloud-based analytics platform for modeling patient's context, personalized care, and recommendations. The development of Cognitive AI health coach is being accelerated for effectively monitoring the patient's health and behavioral information [86], for cognitively processing patient's speech and so on.

User's healthcare data keeps on increasing and it is made available in the cloud for experts analysis. A cognitive technology based data

transmission method [87] has been implemented to analyze the patient's present healthcare data and predict the future treatments. The simulated annealing method and stochastic prophesy is used to predict the future health condition based on the present conditions. The cloud based healthcare with 'h' number of hospitals are defined as $H = \{H_1, H_2, H_3, \dots, H_h\}$, $\forall h \in H$ and $DP = \{DP_1, DP_2, DP_3, \dots, DP_d\}$, $\forall dp \in DP$ represents the number of departments in relation with many hospitals. Let do indicates the number of doctors Do_{ij}^x in i^{th} department of x^{th} hospital and $j = \{1, 2, \dots, do\}$. Let P_{ij}^k indicates the sets of patients in i^{th} department of x^{th} hospital and $j = \{1, 2, \dots, do\}$. Similarly, the data is represented in cloud data centers and analyzed for calculating the data with highest priority representing the patient's who need emergency treatment. Some of the examples of this system are heart ailment analysis and cancer intensity classification. The main issue is that the processing time is very high. So, the future work deals with the development of a data analytical model based on SPARK.

8.6. Cognitive electronic medical records (EMR)

The EMR contains the patient's diagnoses, progress notes, lab & test results, medical histories, demographics, vital signs, and so on. To provide the best treatment to the patients, the medical professionals struggle to make sense of the information available in EMR. Thus, cognitive computing systems are used to transform the data available in EMR to intelligent systems. The Watson patient record summary model [88] is a cognitive EMR system. This model finds the doctor's cognitive needs, visualizes the patient record summary, generates a list of problems, extracts the features using NLP, identifies the feature with the most influential contribution and links it with the medications. An NLP based cognitive decision support system [46] has been designed to automatically detect the status of a disease from the EHR records. Identify the disease conditions of the patients by generating a timeline of disease progression. Watson NLP pipeline takes structured data such as lab values, medical records, and clinical notes as input. The pipeline involves the following steps: disease status evidence extraction, disease treatment goals extraction, feature extraction, lexicon generation, ground truth, and model generation. Regular expression based pattern matching constraints and rule based approaches are developed to determine the disease status. The main issue is the automated label generation and automated feature extraction.

A cognitive computing based application for hearing impaired patients has been developed [89] for automatically extracting meaningful information from the patient's medical records. This system consists of Information Extraction (IE) module and knowledge base module. The knowledge base model deals with two ontological frameworks to categorize the extracted concepts from IE module and maps into medical knowledge base. In future, OntoMathPro software can be used to represent the knowledge of concepts mathematically. The OntoMathPro software is a ontological framework of mathematical concepts and the semantic relationship. The developed knowledge base extracts the evidence based knowledge and identifies the risks, medical problems and so on.

8.7. Cognitive healthcare assistants

Healthcare Cognitive Assistant (HCA) [90] is an intellectual and intelligent assistant which enhances and augments the human intelligence capabilities. These assistants possesses interactive, context awareness and adaptive cognitive abilities. The various application areas are increasing the cognitive ability of people, training clinicians handling inconsistency, uncertainty, situational and contextual awareness. In the field of healthcare, patient's data must be secure and confidential. So, preserving privacy, confidentiality and security is an important concern.

8.8. Personalized medicine

Cognitive Machine Learning (CML) [91] makes use of brain-inspired ML algorithms and fuses the techniques of AI, ML, DL and NLP to intelligently mimic the human thoughts. The survey article focuses on rendering targeted and customized therapy to each person by analyzing the existing issues, pros, cons, applications, and models. The CML techniques analyses the disease status, disease progression and also helps in identifying at risk patients. The process of personalized treatment using CML has been demonstrated. The different processes involved in CML layer are NLP based disease status identification, Artificial Neural Network (ANN) for disease progression, deep reinforcement learning framework, hypothesis and evidence extraction. The CML generates evidence based optimal result. The major issue in personalized treatment is privacy protection and confidentiality. Deep representation in cognitive technology might be a possible solution.

8.9. Intelligent agent-based healthcare

In personalized and socialized medical care platform, agent technologies find out innovative applications with recommendation and decision making abilities. An intelligent agent [92] is a large scale clinical decision support system furnished with expert knowledge to process both structured and unstructured data. The intelligent agent stores and processes the inputs (patient questions), internal process (medical interventions), and outputs (medical treatments). Ambient Intelligence [93] is a promising research area in monitoring the patients and providing individualized support. Personalized medicine delivers services such as prediction, preclusion, treatment of illness, monitoring, and management of health to specific individuals or groups.

Autonomous decision-making agent has been created based on ambient intelligence software and tailored with good learning, reasoning, and pre-planning mechanisms. Watson Discovery Advisor applied to Life Sciences [94] has been developed as a result of a cognitive solution. The four main components are create, compose, comprehend, collect domain based content. This advisor is trained to analyze and understand the technical content by applying predictive modeling, NLP, and ML techniques. In future, the system requires additional augmentation through instance samples.

Temporal modeling [95] in clinical cognitive computing integrates ML classifier based transition models into temporal models like Markov Decision Process (MDP). MDP extends Markov Reward Process (MRP) because it includes decision to be taken by the agent. MDP is a tuple as shown in (15)

$$\langle S, A, P, R, \gamma \rangle \quad (15)$$

where S indicates set of states, A indicates set of actions, P indicates state transition probability matrix, R represents the reward function and γ gives the discount factor. Temporal modeling approach captures certain human cognitive aspects such as perception (observation) and action (treatment) for effective decision making. The major challenge of this approach is treatment adherence variability (medication adherence) and the integration of temporal difference learning into transition models. An innovative agent called incremental type-2 Meta cognitive Extreme Learning Machine (et2ELM) [96] explores the three primary functions of human meta-cognition, such as what to learn, how to learn and when to learn. A learning policy algorithm has been proposed with five stages namely what to learn strategy: active learning strategy, hidden node growing scenario, hidden node pruning mechanism, feature selection mechanism, and adaptation of local sub-model. The output function of et2ELM is shown in Eq. (16)

$$\hat{y}_o = \sum_{i=1}^P h_i^o \left(\tilde{C}_i, \sum_{i=1}^{-1} q_o, X_n \right) x_{ei,o} \quad (16)$$

The detailed architecture has been explained in the article. In future, the design of recurrent type-2 ELM is considered.

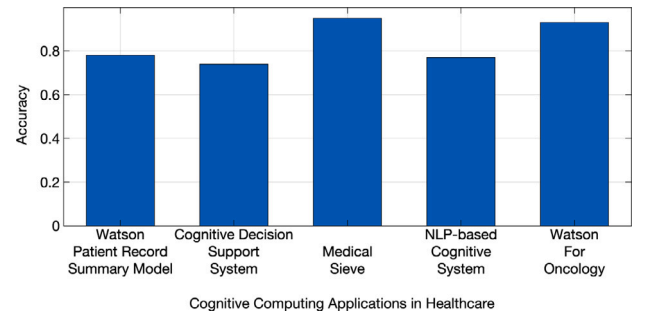


Fig. 13. Accuracy of cognitive systems for implementing cognitive applications.

A bar graph as shown in Fig. 13, explores the accuracy of the cognitive system for implementing cognitive applications. The x-axis indicates the cognitive systems and Y-axis indicates the level of accuracy achieved by those systems. A steady development can be visualized in the graph, and the most successful cognitive system is the Medical Sieve with an accuracy of 0.95. Watson For Oncology cognitive computing system has also achieved a greater precision with an accuracy of 0.93.

Table 10 describes the various cognitive systems which render useful information to the clinicians. The Medical Sieve cognitive system has achieved a high accuracy of 0.95 and thus, serves as a powerful assistant to help radiologists and oncologists. This system integrates the techniques of multimodal analytics, EHR population analytics, and ML. An expert system called cognitive decision support system for prognosis and severity of diseases is the least accurate system with an accuracy of 0.74.

9. Use cases of cognitive computing in healthcare

Cognitive computing systems is a probabilistic approach which learns, reasons, understands from interactions with humans and recommends best actions. These systems make use of computational technologies for data-information-knowledge transformation. The three building blocks of cognitive computing are search (explores relevant data), analyze (unstructured data analysis to discover insights), and interpret (interpret the information with suitable Application Programming Interface (API)). A developed cognitive system must address the cognitive capabilities such as reasoning, planning, perception, problem-solving, decision making, and learning. The use cases of cognitive computing in healthcare, as shown in Fig. 14, focus on user/patient experience, research, clinical application, clinical and operational efficiency.

The mind map contains a detailed analysis of the use cases (green colored boxes). The gray colored boxes represent various capabilities related to a patient-centered approach, personalized treatment, AI-based health-care system, disease risk identification, and large scale analysis of medical data. It also explores about research related cognitive capabilities which involve deep human interaction for continuous learning and reasoning, scaling of expertise by understanding medicine language, empowering cognitive process and operations for knowledge representation, decision making, augmenting exploration and delivery to identify personalized treatment options.

9.1. User/patient experience

Cognitive computing improves the user/patient experience by providing reasonable details about patient health. Cognitive computing provides patient-centric support and warns the patient's about the potential consequences of diseases. The cognitive computing assists in creating an individualized treatment plan [13] to enhance the experience of patient and doctor.

Table 10
Cognitive systems with accuracy.

Year	Cognitive systems	Technique	Evaluation metric
2014	Watson Patient Record Summary Model	Watson analytics-based EMR summarization	Accuracy – 0.78
2016	Cognitive Decision Support System	ML and Fuzzy-Cognitive-Map Mode	Accuracy - 0.74
2016	Medical Sieve	Multimodal, EHR analytics, semantic clinical inference and ML	Accuracy – 0.95
2017	NLP-based Cognitive System	Watson NLP Pipeline, Lexicon generation, Feature Extraction	Accuracy – 0.77
2018	Watson For Oncology (WFO)	AI-based Clinical Decision Support System, NLP and ML	Accuracy – 0.93

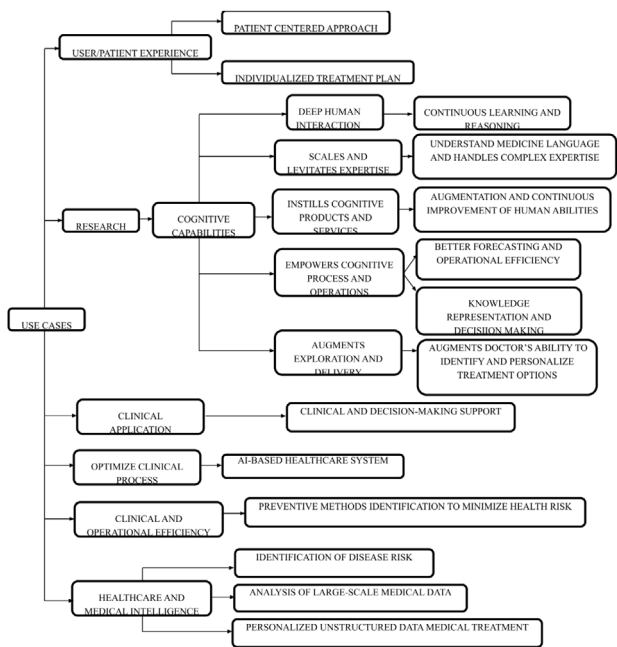


Fig. 14. Use cases of cognitive computing.

9.2. Patient centered approach

In healthcare, cognitive computing systems collect personal, clinical, and social data from a variety of sources to improve patient engagement [52]. The clinician interacts with the model by sifting through the raw data, synthesizing the medical knowledge, and mapping the medical logic from the knowledge to provide cognitive decision support. With the help of cognitive computing, healthcare professionals can access the best viewpoints from internationally known medical experts, travel to remote locations, and save clinical studies to better match patients to care.

9.3. Research

Cognitive computing in healthcare research focuses on processing large volumes of unstructured data, natural language data, medical image data from patients-clinicians interactions and treatments. The cognitive system makes use of computational capabilities and expands the cognitive capabilities to find solutions for real-world problems. The essential cognitive capabilities [18] are deep human interaction, reason at scale, instills cognitive products, empowers cognitive process, augments exploration, clinical application, and optimizes the clinical process.

9.4. Deep human interaction

Cognitive systems create immense human interaction by taking huge data into account from Electronic Medical Records (EMR) and the data from wearables. This system continuously learns, performs reasoning on the structured, unstructured data, and delivers greater insights.

9.5. Scales and levitates expertise

The medical knowledge is expanding at a faster rate because of journals, new practices, and new protocols. In the lifetime, a person generates nearly one million gigabytes of health care data. A cognitive system understands the language of medicine and handles complex expertise.

9.6. Instills cognitive products and services

New products and services with cognition like medical devices are developed to sense, reason, react, and interact naturally with humans. So, this allows for augmentation and continuous improvement of human abilities.

9.7. Empowers cognitive process and operations

Cognitive systems enhance better forecasting, operational efficiency, and decision making. A cognitive medical imaging application analyzes X-rays, Magnetic Resonance Imaging (MRI), and ultrasound images by extracting a deep sense of knowledge that helps in the diagnosis of diseases.

9.8. Augments exploration and discovery

By applying cognitive technologies to huge amounts of data, a holistic approach is developed to create a robust cognitive platform. This platform incorporates NLP, ML, reasoning, human-computer interaction for high-performance computing. This system augments the ability of the doctors to identify and personalize treatment options.

9.9. Clinical application

Cognitive Computing helps us to take the right clinical and decision making support at the right time. The challenging task is to apply the cognitive computing in oncology, pediatrics, and even more specific areas. The real-world problems are solved by cognitive systems using an adaptive approach [16]. This approach performs advanced analysis of data and delivers data-driven insights to the decision-makers.

9.10. Optimize clinical process

The AI-based healthcare system [51] creates a patient map based on health condition for personalized patient care by making use of NLP and ML algorithms. This system is helpful in real-time monitoring of patients vitals by using wireless sensors.

9.11. Clinical and operational efficiency

Nowadays, the most significant amount spent on testing, diagnosis, and treatment for the illness is often a delicate task. So, cognitive computing provides a cost-effective methodology which optimizes procedures and processes.

9.12. Healthcare and medical intelligence

This method identifies the risk of diseases and renders personalized medical treatment to each patient. The most significant resources of data for prognosis and treatment are medical literature, patient's medical records, demographic studies, and research studies. Large scale medical data is analyzed to extract patterns and generate hypotheses. Thus, the cognitive system provides a platform [16] for scientists to conduct various experiments, verify the hypotheses, and feeds it back to generate new hypotheses.

Thus, use cases analyze the various features related to a particular goal. Use cases improve user/patient experience for supporting the patient-centered approach, optimizes the clinical process for developing an efficient decision-making model.

9.13. Case studies of cognitive computing in healthcare

Several case studies of cognitive computing in healthcare are illustrated below:

9.13.1. Case 1: A multidisciplinary panel of experts on tumours

Personalized care for breast cancer patients [62] is challenging with rapidly changing scientific evidence, drug approvals, and treatment guidelines. The tumor board's treatment recommendations were considered concordant if WFO designated them as 'recommended' or 'for consideration'. According to stage and patient age, concordance rates varied. WFO was reliable in this application, despite the fact that drug therapy accessibility, professional treatment guidelines, and the expertise of experts who trained the computing system will all affect concordance.

9.13.2. Case 2: Disease status identification using NLP-based cognitive system

A cognitive decision support system [46] that uses natural language processing (NLP) to automatically determine the state of a disease from the clinical notes in a patient record. The system uses supervised or semi-supervised learning methods and IBM Watson Patient Record NLP analytics. It makes use of unstructured text from clinical notes, information from a patient record's structured section, and disease control objectives from clinical guidelines. The experimental findings demonstrate that the suggested model can identify disease status passages with an average F1-score of 0.86 and classify the status as controlled or uncontrolled with an average accuracy of 0.77 using supervised learning techniques.

9.13.3. Case 3: Colon cancer treatment based on cognitive computing decision support system

A clinical decision-support computing system called IBM Watson for Oncology [44] offers oncologists evidence-based treatment recommendations for a range of cancer diagnoses. The recommendations for evidence-based, supported treatments are divided into three groups: recommended, representing the preferred strategy of the Memorial Sloan Kettering Cancer Center (MSKCC). A significant advancement in oncology is cognitive computing, which has the potential to greatly enhance the calibre of medical decision-making.

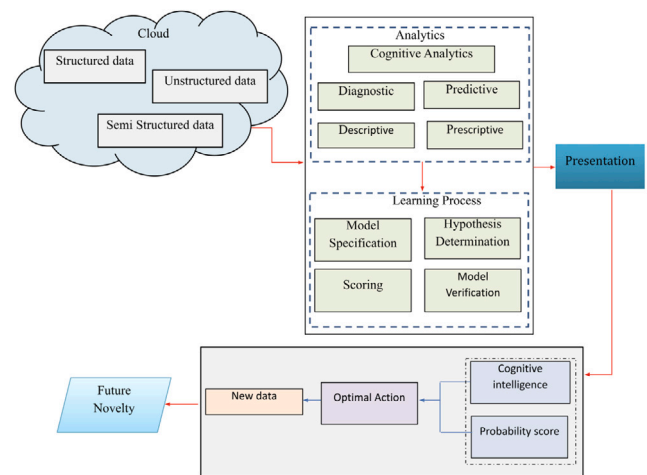


Fig. 15. Conceptual model of cognitive computing.

9.14. Conceptual model of cognitive computing

A conceptual model of cognitive computing [43] has been analyzed as shown in Fig. 15. The model takes the healthcare cloud data such as structured, unstructured and semi-structured data as input. The cognitive computing platform consists of analytical services, continuous learning process, presentation and visualization. The analytical services include descriptive, diagnostics, predictive, prescriptive and cognitive. The output of analytical service is best and optimal action. The continuous learning process starts with model specification, hypothesis determination, scoring and model verification. The presentation and visualization section deals with knowledge creation which involves insights generation and optimal decision making.

10. Discussion: Past, present and future of cognitive computing

The cognitive computing systems automatically extract the knowledge from the data. In the future, these systems would be designed to generate wisdom from the acquired knowledge automatically. The transition starts from traditional to smart and then to cognitive healthcare [56]. The demand for cognitive computing technologies in healthcare domain is very high because of the uncontrollable data flow.

Cognitive computing gives incogitable benefits to people in leading a healthier lifestyle. The time required by the medical researchers in formulating a treatment plan for the patients is extensive. Cognitive computing in healthcare scales, augments, accelerates the expertise of medical professionals and equips them to handle next-generation healthcare data effectively using cognitive technologies. Cognitive computing also offers an evidence based clinical decision system. Sifting through the large unstructured scale medical data is a tedious process, so cognitive computing technology processes the data using NLP discovers patterns and valuable insights.

WatsonPaths [71] is a scenario based cognitive inference generation system which serves as an automated question answering system to render best treatment inferences by analyzing the patient's condition. This cognitive system performs scenario (input question) analysis, node prioritization, relation determination, belief calculation, hypothesis formulation, and confidence refinement. The main challenge in this system is that even though the system retrieves the accurate answer there is no explanation on how and why the answer is correct.

The major challenge lies in the privacy of patient data so the cognitive systems must secure the patient's data by providing authentication. The another important issue to be noted is that these cognitive computing systems are still in its developmental stage and not fully deployed and maintained. Some of the other issues are lack of hands

Table 11
Highlights of three most competent cognitive systems.

Cognitive systems	Products	Key features	Applications	Use cases — Research
IBM Watson	IBM Watson Health	Cognitive Healthcare Solutions – leverage data, enable AI, optimize collaboration	Watson for Oncology, drug discovery, genomics, imaging clinical review, imaging patient synopsis, Clinical trial matching	Drug discovery — Parkinson's disease, population health, clinical operations, clinical trials coordinator
Google Deepmind	Deepmind health	Uncovering new knowledge, energy efficiency, data center	Continuous prediction of future acute kidney injury, digitally enabled care pathway, diagnosis of eye diseases	Transformation of drug invention, Preventive treatment for life threatening illness
Microsoft Cortana	Cortana Intelligence Suite	Personalized patient care, optimize clinical and operational effectiveness	Early detection of breast cancer, predicts medical emergencies, eye care	Microsoft's intelligent personal assistant — healthcare bot, Integrate advanced AI technologies into Cortana

on experience, construction of knowledge base, training the cognitive system and so on. There is also some kind of discrepancy between machines work and human's work which is noted in IBM Watson when applied for cancer treatments [97]. IBM Watson's thinking goes in a statistical way and identifies the basic outcomes whereas doctors think in a different way. Mining the useful information from patient's EMR is also a major issue. IBM Watson lacks in dynamicity and there is no comparison of the new patient with the existing patients. So, these cognitive systems need to be implemented in the research institutions in order to overcome the above mentioned issues.

Table 11 presents the highlighted features of eminent cognitive systems. These three efficient cognitive systems plays a significant role in healthcare. From these valuable insights, medical professionals can easily find solutions and treatments for many life-threatening diseases. Thus, cognitive computing technology reduces diagnostic and therapeutic errors in clinical practice.

Some other challenges or limitations of cognitive systems [3] are

- Risk analysis
- Lack of training and more usage of AI algorithms
- Security issue due to huge amounts of unstructured information
- Long time for development

To summarize, these cognitive systems provides optimal patient care, cost-effective and personalized treatment. Cognitive computing systems renders best evidence based treatment plans by infusing, analyzing and processing large volumes of data.

10.1. Cognitive computing challenges and future research directions in healthcare

Application of Cognitive Computing techniques is an important research topic in ML and AI society. The novelty of this article is that it focuses on the present issues and challenges of cognitive computing in healthcare. The mindmap in Fig. 16 portrays the present issues and future enhancements of Cognitive Computing in healthcare.

The light yellow colored boxes represent the present issues, and the deep yellow colored boxes represent the future enhancements for the issues. To highlight some of the future enhancements are deep cognition, precision medicine, cognitive handwriting recognition system, disease progression modeling, disease status identification, automated patient, monitoring, recommendation, and coaching.

Table 12 renders certain future enhancements related to healthcare.

10.1.1. Doctor's complex notes

Doctor's medical notes contain complex medical jargons, and it is complicated to extract useful and meaningful information from these notes. So, Microsoft cognitive service is employed to the concept of handwriting recognition and detection.

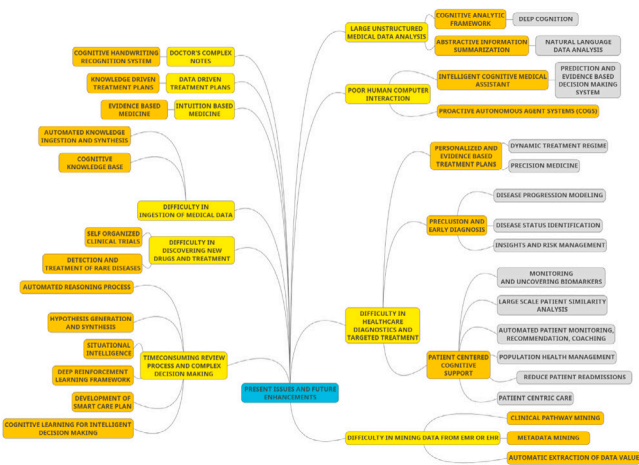


Fig. 16. Present issues and future enhancements of cognitive computing in healthcare. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

10.1.2. Data-driven treatment plans

Data-driven treatment plans lack evidence-based decision-making capability, so expert knowledge-driven treatment plans are at an evolution stage, to provide evidence-based decisions. In some cases, a systematic approach combining both data-driven and knowledge-driven treatment plans has been used to provide effective and personalized treatment.

10.1.3. Intuition based medicine

Practitioners can experience intuition-based medicine, but it is inaccessible to novice users. Evidence-based medicine is accessible to all practitioners as well as novice users. Evidence-based medicine can be figured out as a four-step framework [98], which includes frame, find, evaluate, and apply.

10.1.4. Ingestion of medical data

Ingestion deals with the storage and retrieval of medical data. Ingestion of medical data is a time-consuming process, and it is tough to manage the data. So, the critical aspect of cognitive computing called automated ingestion and synthesis of knowledge by the development of CKB [59] emerged to overcome the problem of ingestion.

10.1.5. Discovery of new drugs and treatment

Discovery of new drugs and treatment is a challenging problem to address. Drug discovery opens up a path for new drugs to be discovered. Self-organized clinical trial aims at providing new treatment, different combinations of treatment, new drugs for the existing disease, and new insights into a disease.

Table 12
Present issues and future enhancements.

Present issues	Future enhancements
Doctor's complex notes	Cognitive Handwriting Recognition System
Data-driven treatment plans	Knowledge-driven treatment plans
Intuition based medicine	Evidence-based medicine
Ingestion of medical data	Automated knowledge ingestion, Cognitive Knowledge Base (CKB)
Discovery of new drugs and treatment	Self-organized clinical trials
Time-consuming review process	Cognitive Assistant with automated reasoning, hypothesis and evidence generation
Complex decision making	Cognitive learning for intelligent decision making
Large unstructured medical data analysis	Cognitive Analytic framework and deep cognition
Poor human-computer interaction	Intelligent Cognitive Medical Assistant and proactive autonomous agent systems
Healthcare diagnosis	Personalized Dynamic Treatment Regime
Targeted treatment	Patient-centered cognitive support
Mining data from EMR or EHR	Clinical pathway and metadata mining

10.1.6. Time-consuming review process

The review process is too complex and time-consuming. Patients need to wait for more than a week to get their reports. So, a cognitive assistant is developed with cognitive computing techniques such as automated reasoning, hypothesis, and evidence generation. This intelligent cognitive assistant can provide a cognitive summary of patient records and make the review process easier.

10.1.7. Complex decision making

Medical decision making is quite complex for serious and chronic diseases because the doctors must review a significant amount of past medical records of the patients. So, the integration of cognitive learning technique in AI-based analysis helps in intelligent medical decision making. The cognitive medical system [99] assists in the diagnosis and helps the doctors to take correct and complex decisions.

10.1.8. Large unstructured medical data analysis

Cognitive Analytic framework [5] and deep cognition are employed to process a large amount of unstructured medical data. The cognitive analytic framework fuses the techniques of knowledge representation, inference, reasoning, ML, learning and adaptation, hypothesis generation, and validation.

10.1.9. Poor human-computer interaction

Cognitive systems learn from data, builds knowledge over the data, understands the natural language, reasons with purpose, and interacts with humans. Development of Intelligent Cognitive Medical Assistant and proactive autonomous agent systems facilitates natural interaction with humans.

10.1.10. Healthcare diagnosis

The medical field is facing the issue of correct healthcare diagnosis. Cognitive computing technology helps in the right diagnosis of health-

care and proves that personalized Dynamic Treatment Regime (DTR) is best for providing optimal treatment. Personalized DTR fuses ML, AI technologies, and provides optimal treatment to the right people at the right time.

10.1.11. Targeted treatment

Cognitive computing provides patient-centered cognitive support. The patients have full control of their health in a personalized way [100]. Cognitive computing enhances the clinicians diagnostic accuracy and processes their presaging prediction. Some cognitive tool developed by IBM Watson Health such as Watson for Oncology [101] provides targeted treatment for cancer patients.

10.1.12. Mining data from EMR

Difficulties in the use of EMR data includes noisy data, incomplete and inconsistent data. The Watson patient record summary model [88] serves as a cognitive EMR system. This model finds the doctor's cognitive needs, visualizes the patient record summary, generates a list of problems, extracts the features using NLP, identifies the feature with the most influential contribution and links it with the medications. Clinical pathway and metadata mining can be employed in cognitive EMR system. In future, the cognitive EMR system can be employed for patient status summary extraction, natural language understanding and cognitive prognosis. Thus, Cognitive Computing blends the technologies of AI, ML, and NLP to determine the mode of disease and to provide optimal personalized treatment. Personalized model is an efficient model which processes the medical records with higher precision value. Personalized medicine is an optimal pathway [102] for taking individualized health decisions. The personalized model cares the patients with a specific and unique condition by rendering patient centric care.

10.2. Recent applications and research directions of cognitive computing on COVID-19

The growth of digital technologies such as cognitive computing fusing AI and ML proves to be very effective in processing the medical data and rendering the evidence based decisions. A federated deep learning based COVID-19 prognosis and classification model has been proposed by using the IoT technologies [103]. Patient's data is recorded by the IoT devices and SqueezeNet deep learning architecture is used for analysis of Chest X-ray (CXR) images. The objective function for the SqueezeNet architecture noted as f_u is defined as shown in Eq. (17)

$$\arg \min_{\Theta^u} L = \sum_{i=1}^n l(y_i^u, f_u(x_i^u)) \quad (17)$$

An optimization algorithm is also employed to tune the parameters. The major challenge noted is collection of training data. In future, data offloading and resource management approaches can be focused.

The cognitive technologies are useful for diagnosing, monitoring, treating, researching and preventing COVID-19. For diagnosing the disease, self evaluated cognitive agents can be focused to identify the symptoms and accelerate the diagnosis process. For monitoring the disease, cognitive systems can be designed in such a way those systems systematically observe and analyze the vital parameters. For treating the disease, cognitive chatbots with reasoning and understanding capability can also be developed and enriched with medical knowledge so that it serves as a medical prodigy for doctors and patients. Research is carried out on drug discovery, clinical trial matching, information extraction for combating COVID-19. For disease prevention, cognitive technologies can be developed to give warnings or signals with necessary precautionary instructions.

10.3. Impact analysis of cognitive computing in healthcare

Cognitive computing provides intuitive and interactive computing that expands knowledge of the clinical domain and enables informed decision-making about the most appropriate treatment for diseases using relevant information provided by analytic tools.

- Analyses genomic data to assist researchers.
- Contributes to the development of personalized medicine and clinical decision-making.
- Improves communication in care settings by converting unstructured documents into structured data.
- Assesses the health needs of patients and the cost of care.
- Make individual insights available in a timely manner.
- Analyses the probability of success and recommends interventions based on that information.

10.3.1. Cultural change of cognitive computing in healthcare

In healthcare, cognitive computing combines the functions of humans and machines together to improve human decision-making. The use of cognitive computing has the potential to facilitate a cultural change in the practice of medicine. This will enable physicians to provide intelligent recommendations to patients on how to treat themselves. Additionally, cognitive computing aids in the creation of individual treatment plans, improving both patient and doctor experiences. In images, it enables radiologists to find anomalies more quickly.

10.3.2. Outlook of cognitive computing

In healthcare, cognitive computing is primarily used by providers. Globally, providers hold a 44.5% market share. In real time, cognitive computing can analyze emerging patterns, spot business opportunities, and handle critical process-centric issues. As a result of examining a vast amount of data, cognitive computing systems like Watson can simplify processes, reduce risk, and pivot as the situation changes.

11. Conclusion

The ultimate aim of cognitive computing is to enable a machine to possess a brain like cognitive intelligence. In other words, the goal of cognitive computing is to build systems that can learn and naturally interact with humans. The power of cognitive computing systems lies in the informed decisions taken by humans with the help of machine intelligence. Since cognitive computing transforms the way we live, work, and think it is called the future of intelligence. Healthcare-related cognitive computing technology analyzes, understands all the structured, unstructured information related to the patient's condition and provides more information for the doctors on the treatment by giving exhaustive output about their health. Cognitive computing technology is not a supplement for doctors but it serves as an assistant which can give efficient treatment options within shorter time period. This technology offers real-time medical insights to the clinicians and expands their knowledge in designing personalized medicine. Thus, cognitive computing technology helps the medical professionals to make an informed, timely decision and deliver the best solution to the patients. This article concludes that cognitive computing technology does not supersede the doctors in the medical industry but provides various optimal decisions or treatment options within short period of time.

11.1. Limitations

The limitation of cognitive computing is i. In cognitive computing, digital devices manage critical information, so security is a major concern. ii. Developing cognitive computing products and systems takes a considerable amount of time. iii. While cognitive computing systems complement individual intelligence and analysis, they rely on humans to make decisions and lack an automated decision-making process. iv. Adoption rates are slow due to the long development lifecycle.

11.2. In scope of cognitive computing

Engagement: Data repositories of cognitive systems contain both structured and unstructured information. Experts can develop deep domain insights and provide assistance.

Decision: Cognitive systems' decisions are constantly changing in response to new data, results, and actions. The ability to determine why a specific decision was made and alter the confidence level of a system response are prerequisites for autonomous decision-making. IBM Watson's application in the field of healthcare is a common use case for this model.

Discovery: In cognitive computing, discovery represents the highest level of sophistication. Discovering insights and understanding vast amounts of information requires developing skills.

11.3. Out scope of cognitive computing

The beyond scope of cognitive computing is connected healthcare. The term "connected health" refers to a socio-technical framework for the administration and delivery of healthcare that makes use of technology to deliver healthcare services remotely.

11.4. Future trends

The main goal of cognitive computing technology is to develop self-working and learning systems for handling complex problems. The cognitive cycle makes use of existing information to generate innovative information. Moreover the cognitive processing framework, integrates the information from different sources and renders the optimal answers. In future, the cognitive systems must focus more on privacy, security, hands on experience, knowledge base construction, training, dynamicity, informative thinking, and so on. The emerging innovations and future applications are deep question answering, hypothesis generation, neuro-symbolic AI, evidence-based reasoning, data integration, data representation, quantum computing using cognitive technology. These systems also serves as a pathway between patients and healthcare providers by enhancing the interaction. Many highlighted features of cognitive computing systems are cost-effective, patient-centric technology, context switching, personalized or customized technology, ability to learn from past experiences, extension with new capabilities, anticipate and act according to the user's needs and interest. The other futuristic trends of cognitive computing in healthcare are customized knowledge base medicine, better disease preclusion and prognosis, epidemic prevention, optimal treatment decision, and cognitive intelligence.

CRediT authorship contribution statement

Srivani M.: Contributed in collecting research articles, Surveying, Designing, Paper writing. **Abirami Murugappan:** Proposed ideas, Guidance on paper writing. **Mala T.:** Proposed ideas, Guidance on paper writing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent

Informed consent is not required because no human participants or animals are involved.

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